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Meshfree simulation of metal cutting: an updated Lagrangian approach with dynamic refinement

M. Afrasiabi [a,b,*], M. Roethlin [b], H. Klippel [b], K. Wegener [b]

[a] Chair of Structural Mechanics & Monitoring, D-BAUG, ETH Zürich Stefano-Franscini-Platz 5, Zürich 8093, Switzerland
[b] Institute of Machine Tools & Manufacturing (IWF), D-MAVT, ETH Zürich Leonhardstrasse 21, Zürich 8092, Switzerland

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ABSTRACT

This article presents an implementation of a meshfree method, intended for the application to simulate an orthogonal metal cutting operation. To optimize the runtime, a dynamic refinement algorithm via particle splitting is also employed in the present work. The workpiece material is made of TiAl6V4, for which a modified Johnson-Cook constitutive modeling is incorporated from the state of the art. This flow rule is chosen to include the strain softening phenomenon resulting from damage in the machining of titanium alloys. The numerical analysis is thermo-mechanically coupled and formulated in the updated Lagrangian framework. Following a preliminary investigation of the refinement algorithm, the metal cutting process is simulated with four models and cross-compared to assess the workability and efficacy of the proposed developments. In the interest of reproducibility, the code is made open-source and can be downloaded from https://github.com/mroethli/mfree_iwf-ul-cut-refine.

1. Introduction

Modeling of metal cutting processes is a fundamental component of today's manufacturing advances. Process variables that are not feasible to measure experimentally in metal cutting can be observed with the help of numerical models. These models can also survey the parameter space more comprehensively, providing a basis for the optimization of cutting conditions, tool geometries, and other parameters. However, a recurring question arises: what is an efficient way to do so? One efficient way to do so is to use meshfree methods.

Introduced by Gingold et al. [1] and Lucy [2] in 1977, Smoothed Particle Hydrodynamics (SPH) is still the most popular meshfree method, as evidenced by a myriad of interesting applications in various fields from solid [3,4] to fluid [5–10] mechanics problems, to just name very few. Using SPH, the governing Partial Differential Equations (PDEs) are discretized in space by employing a set of particles. These particles possess material properties and interact with each other in a fashion controlled by a smoothing kernel function. The main advantage of this meshfree type of discretization in metal cutting simulations is its ability to naturally handle large deformations with no theoretical limit and without the caveat of mesh distortion (since there is no mesh). Some successful works in this regard include [11–14]. This methodology is in contrast to similar simulations using the Finite Element Method (FEM) for which handling mesh distortion in large deformations requires extra treatments for the stability of the numerical solution, and subsequently mesh re-computation (see [15–17], for example).

On the other hand, it is understood from a literature review that most of the meshfree metal cutting simulations are performed with particles of uniform spacing, leading to a single-resolution configuration. Adaptivity, or more specifically variable resolution, can be utilized to improve the efficiency of numerical simulations without reducing their accuracy. This is a profitable approach in contemporary meshfree methods, especially for problems where a significant change in the values of their state variables is restricted to small areas of the domain (i.e., the critical sub-domains). Needless to say, a finer discretization for these particular regions is desired. Excellent developments can be found in the state of the art with several applications including the shock tests [18,19], the vortex methods [20,21] and advection-diffusion simulations [22], fluid and multiphase flows [23–28], and the works of [4,29,30] for 3D problems.

Due to the same phenomenological perception, it turns out that metal cutting processes would also stand amongst the most fitting candidates for adaptivity. There is, nevertheless, a surprisingly low number of accounts that adopt a multi-resolution approach in meshfree metal cutting simulations. About 10 years ago, Rabczuk and Samaniego [31] modeled cutting of a 3D block with particles using a plasticity material model

Abbreviations: JC, Johnson-Cook; MJC, Modified Johnson-Cook; SHPB, Split-Hopkinson Pressure Bar; CFD, Computational Fluid Dynamics; CFL, Courant-Friedrichs-Lewy; PDE, Partial Differential Equation; FEM, Finite Element Method; SPH, Smoothed Particle Hydrodynamics; PSE, Particle Strength Exchange; CSPM, Corrective Smoothed Particle Method; ICSPM, Improved CSPM; FNMS, Fatehi's New Meshfree Scheme.

* Corresponding author.
E-mail address: afrasiabi@ethz.ch (M. Afrasiabi).

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Nomenclature

χ	Taylor-Quinney coefficient
Δx	Particle spacing
κ	Dispersion coefficient of “child” particles
ρ	Density
σ_y	Yield stress
$\underset{\approx}{\boldsymbol{C}}$	Elastic moduli tensor
$\underline{x}$	Position vector
$\underline{\underline{\sigma}}$	Cauchy stress tensor
$\underline{\underline{\epsilon}}$	Strain tensor
$\underline{\underline{D}}$	Rate of deformation tensor
$\underline{\underline{K}}$	Heat conductivity tensor
$\underline{\underline{L}}$	Velocity gradient
$\underline{\underline{S}}$	Deviatoric stress tensor
ξ	Smoothing ratio of “child” particles
c_p	Specific heat capacity
h	Smoothing length
k	Heat conductivity
N	Total number of particles
n_c	Normalization factor
T	Current temperature
t	Time
T_0	Initial temperature
T_m	Melting temperature
W	Smoothing kernel
x, y	Spatial coordinates

with an isotropic hardening/softening law. In their work, however, a reference result from either FEM simulations or experimental data is lacking and only a qualitative illustration is presented. In 2014, Spreng et al. [32] simulated an elementary 2D orthogonal cutting process with SPH. Both refining and coarsening procedures were employed within the course of spatial adaptation and consequently applied to their cutting test. While suffering from the same soar point as [31] in terms of results validation, the spatial coherence and quality of the chip formation in [32] are notably worse than their corresponding standard SPH single-resolution reference. In their more recent work, nonetheless, Spreng and Eberhard [33] verified the adaptive-resolution SPH result by comparing it to the experimental data. For this purpose, they took the cutting force of a 3D oblique cutting test into account. In a nutshell, the improvement gained by adaptivity is not readily recognizable, neither from a quality/quantity perspective nor from a computational cost saving viewpoint.

Upon the previous observation, one notices that meshfree simulation of metal cutting processes is not really mature when comparing it to other fields of application. To partially fill this gap the present work seeks to accelerate and increase the size of simulations for an orthogonal metal cutting process by incorporating dynamic refinement into the spatial discretization of the PDEs. Additionally, a modified version of the standard Johnson-Cook material model as presented by Sima and Özel [34] is implemented in order to include damage resulting from the strain softening phenomenon in the machining of titanium alloys. This issue cannot be addressed when using the standard Johnson-Cook model in a thermo-mechanical analysis of TiAl6V4 cutting [14]. The robustness of the meshfree method is ensured by adopting a set of stabilization terms, and further enhanced by exploiting a linear-complete (i.e., first-order consistent) scheme for comparison purposes. Not only can a more realistic chip morphology be captured as a result of this contribution, but also the solver performs almost 3 times faster than single-resolution models thanks to the efficiency proposed by dynamic refinement. Furthermore, the cutting model considered herein is thermo-mechanically coupled, since the metal cutting simulations without taking thermal effects into account, like the results presented by Rabczuk and Samaniego [31] and Spreng et al. [32], can yield unrealistic results. Aligned with the principle of “rising complexity”, this work strives to pave the road towards a thermo-mechanically coupled, optimally adaptive, meshfree, software tool for the simulation of metal cutting processes.

This paper is structured as follows: in Section 2 the requirements for understanding the metal cutting model is presented with the system of governing equations in the updated Lagrangian framework. This section presents basic terminology and background that sets the scene for the following material. Section 3 outlines the stabilized SPH formulation of the model. This is followed by Section 4, where the spatial refinement algorithm is expressed. A prerequisite study is then investigated on the analysis of the density refinement errors in Section 5. Next, the results of metal cutting simulations are discussed within Section 6, and compared against reference data taken from the literature. In Section 7, the paper closes by summarizing the added values offered through this work, and proposing a way forward for better meshfree metal cutting simulations.

2. Metal cutting model

This section concerns with a brief description of the problem at hand followed by the associated mathematical formalism. Firstly, the process itself is described with the help of two schematic illustrations. Secondly, the physical model is formulated without detailing the derivations. The numerical method to be discussed in Section 3 is to be applied for discretizing the governing equations outlined in this section, i.e., Eqs. (1)–(5).

2.1. Problem statement

The metal cutting process in this study is simplified to an orthogonal cutting case. The “orthogonal” simplification allows for consideration of the process in 2D, forming the basis of most analytical and numerical models in this context. This assumption is due to the fact that a sufficiently large ratio of the width over the depth of cut can reduce the strain state of the workpiece to a 2D problem [35], hence the plane strain condition. Fig. 1 illustrates the essential terminology and areas of an orthogonal metal cutting operation, according to the standard textbooks such as [36] and [37]. In this figure, the basic definitions related to the tool and cutting geometry such as the rake angle γ and clearance angle α (i.e., the inclination between the flank face and the new surface), the cutting depth (also known as, feed) are demonstrated. Similar to the majority of the former works in this area, the tool is herein assumed to be a rigid body cutting through a workpiece made of the most frequently used titanium alloy, TiAl6V4.

To derive a good metal cutting model, understanding the chip formation mechanism is as crucial as the reliability of the process parameters.

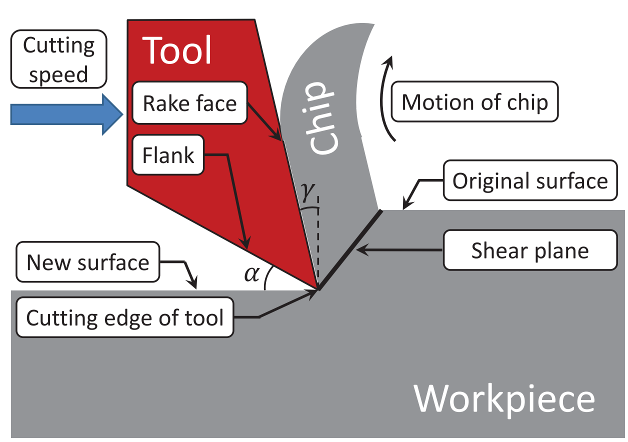

Fig. 1. Schematic of an orthogonal metal cutting process.

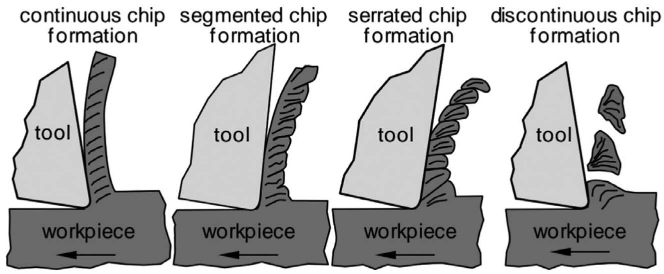

Fig. 2. Different chip formation mechanisms in metal cutting from [38].

A general classification can categorize the chip formation mechanism into the four types distinguished in Fig. 2.

Experimental studies show that TiAl6V4, or titanium alloys in general, belong to the group of materials that generate serrated chips under a wide variety of cutting conditions [38,39]. When simulating a metal cutting process, the predicted morphology of the chip is strictly influenced by phenomenological, e.g., the material model, as well as methodological, e.g., the accuracy of the numerical method, aspects. While the methodological viewpoint is the main concentration of this study, the phenomenological one should not be underrated. This would, therefore, necessitate a material model for obtaining a more realistic chip formation. A suitable choice for this process shall contain a system of equations that captures the forces (or stresses), the associated deformations, and the thermal aspects within a realistic definition of the material behavior.

2.2. Governing equations

The mathematical problem is described by continuum-mechanical equilibrium condition and its associated boundary conditions as well as the Fourier heat equation with boundary conditions, both according to Fig. 1. In addition, as for the constitutive equations, a flow and a hardening rule to describe the plastic deformation, and Hooke's law for the elastic deformation, are formulated. Therefore, one needs to solve the following equations simultaneously to obtain a thermo-mechanically coupled model in the updated Lagrangian framework. This expression follows the procedure given in [40]. The governing system consists of the three conservation laws (i.e., mass, momentum, and energy) for solid mechanics expressed in Eqs. (1)–(3), plus the constitutive equation in Eq. (4) and the kinematic equation in Eq. (5), written altogether as

$$\dot{\rho} = -\rho \nabla \cdot \underline{v} \tag{1}$$

$$\underline{\dot{v}} = \frac{1}{\rho} \nabla \cdot \underline{\underline{\sigma}} + \frac{1}{m} \underline{b} \tag{2}$$

$$\dot{e} = \chi \frac{1}{\rho} \left(\underline{\underline{\sigma}} : \underline{\underline{\dot{\varepsilon}}}^{pl} \right) + \frac{k}{\rho} \nabla^2 T \tag{3}$$

$$\underline{\underline{\dot{\sigma}}} = f(\underline{\underline{\sigma}}, T, \cdots) \tag{4}$$

$$\underline{\dot{x}} = \underline{v} \tag{5}$$

where ρ is the density, m the mass, $\underline{v}$ the velocity, $\underline{\underline{\sigma}}$ the Cauchy stress tensor, $\underline{b}$ the body forces, e the specific internal energy, χ the fraction of plastic work converted into heat (i.e., Taylor-Quinney coefficient [41]). In Eq. (3), $\underline{\underline{\dot{\varepsilon}}}^{pl}$ denotes the plastic strain rate, which is the symmetric part of the velocity gradient restricted to the plastic part. Also, k and T are the heat conductivity and temperature, respectively, assuming the heat conduction to be isotropic ($\underline{\underline{K}} = k\underline{\underline{I}}$). In fact, this energy equation states that the specific energy is stored as a combination of the thermal and mechanical aspects without taking external heat sources into account. In addition to the parameters defined for the conservation equations, $f(\cdots)$ designates a proper constitutive equation to be used for the problem at hand, and $\underline{x}$ represents the position vector of a material element. Besides, a co-rotational time derivative of $\underline{\underline{\sigma}}$ is used in lieu of $\underline{\underline{\dot{\sigma}}}$ which will be described in the remainder of this section. The conventions used for more readability throughout this paper are

$$\begin{cases} \langle\diamond\rangle & \text{approximate value of } \diamond \text{ using numerical methods} \\ \dot{\diamond} & \text{time derivative of } \diamond \text{ (i.e., Newton's notation)} \\ \diamond & \text{if } \diamond \text{ is a zeroth-order tensor (i.e., scalar)} \\ \underline{\diamond} & \text{if } \diamond \text{ is a first-order tensor (i.e., vector)} \\ \underline{\underline{\diamond}} & \text{if } \diamond \text{ is a second-order tensor} \\ \underset{\approx}{\diamond} & \text{if } \diamond \text{ is a fourth-order tensor} \end{cases} \tag{6}$$

2.2.1. Contact forces

Due to the contact between the tool and the workpiece, contact forces should be taken into account to preserve the momentum balance in Eq. (2). This is typically done by associating the boundary conditions in Eq. (2) with the corresponding contact forces. Along the contact zone, the following simple, but standard, Coulomb's friction model is applied

$$\left| \underline{f}^{\,\text{fric}} \right| = \mu \left| \underline{f}^{\,\text{cont}} \right| \tag{7}$$

with a constant coefficient of friction μ, the contact force $\underline{f}^{\text{cont}}$ which is normal to the contacting surface, and the friction force $\underline{f}^{\text{fric}}$ which acts on the contact tangential plane. Given a rigid cutting tool, a penalty type contact force is meanwhile considered as successfully employed by many researches in the field (see e.g., [42–44]). The detailed description of the contact algorithm as well as the parametric tool design can be found in [45], and is not presented again in this paper.

2.2.2. Energy terms & thermal aspects

The term e in Eq. (3) can be substituted with $e = c_p T$, where the specific heat capacity is denoted by c_p. As a result, Eq. (3) is rewritten as

$$\dot{T} = \underbrace{\frac{\chi}{\rho c_p} \left(\underline{\underline{\sigma}} : \underline{\underline{\dot{\varepsilon}}}^{pl} \right)}_{\text{dissipation}} + \underbrace{\frac{k}{\rho c_p} \nabla^2 T}_{\text{conduction}} \tag{8}$$

in which $\underline{\underline{\dot{\varepsilon}}}^{pl}$ was already defined and the dissipation term due to plastic work is computed from [11,41]

$$\dot{T}_{\text{dissipation}} = \chi \frac{\sigma_y \dot{\bar{\varepsilon}}^{pl}}{\rho c_p} \tag{9}$$

and the conduction term needs to be discretized in space using the desired numerical scheme. $\dot{\bar{\varepsilon}}^{pl}$ is the equivalent plastic strain rate and σ_y is the effective yield stress.

2.2.3. Constitutive modeling

As mentioned earlier, $f(\cdots)$ in Eq. (4) corresponds to the material model chosen for evolving the hardening state. A set of coupled constitutive equations are presented in the notion of small elastic strains and extension to the large plastic strains framework. In the elastic domain, the relation simply reads

$$\underline{\underline{\sigma}} = \underset{\approx}{C} : \underline{\underline{\varepsilon}} \tag{10}$$

$$\underline{\underline{\dot{\sigma}}} = \underset{\approx}{C} : \underline{\underline{\dot{\varepsilon}}} \tag{11}$$

with $\underset{\approx}{C}$ and $\underline{\underline{\varepsilon}}$ indicating the (fourth-order) tensor of elastic moduli and the (second-order) strain tensor, respectively. In addition, decomposing the Cauchy stress tensor as the sum of hydrostatic and deviatoric parts yields

$$\underline{\underline{\sigma}} = \underbrace{\underline{\underline{S}}}_{\text{deviatoric}} - \underbrace{p\underline{\underline{I}}}_{\text{hydrostatic}} \tag{12}$$

Since the plastic deformation is volume preserving and thus the hydrostatic pressure is dependent on the Hooke's law only, the pressure p in the hydrostatic part needs to be computed from a suitable equation of state. Among a host of available options for different applications in the literature, the equation below is used according to Libersky and Petschek [46], who applied it to SPH

$$p = c_0^2(\rho - \rho_0) \tag{13}$$

if c_0 is the reference speed of sound. Care has yet to be taken as the rate of the Cauchy stress in Eq. (11) is not indifferent to rigid body rotations, meaning that spurious (dummy) stresses are produced by rotation. As originally derived by Jaumann [47] in 1911, choosing a co-rotating time derivative of $\underline{\underline{S}}$ would be the other alternative. The Jaumann rate of the deviatoric stress tensor calculates

$$\dot{\underline{\underline{S}}} = 2G\left(\dot{\underline{\underline{\epsilon}}} - \frac{tr(\dot{\underline{\underline{\epsilon}}})}{3}\underline{\underline{I}}\right) - \underline{\underline{S}}\,\underline{\underline{\omega}} + \underline{\underline{\omega}}\,\underline{\underline{S}} \tag{14}$$

$$\underline{\underline{D}} = \frac{1}{2}\left(\underline{\underline{L}} + \underline{\underline{L}}^{\intercal}\right) \tag{15}$$

$$\underline{\underline{\omega}} = \frac{1}{2}\left(\underline{\underline{L}} - \underline{\underline{L}}^{\intercal}\right) \tag{16}$$

$$\underline{\underline{L}} = \nabla \otimes \underline{v} = \underline{\underline{D}} + \underline{\underline{\omega}} \tag{17}$$

in which G is the shear modulus, $\dot{\underline{\underline{\epsilon}}}$ the strain rate tensor, and $\underline{\underline{D}}$ represents the symmetric part of the velocity gradient $\underline{\underline{L}}$. The spin tensor is denoted by $\underline{\underline{\omega}}$ which is the skew-symmetric part of $\underline{\underline{L}}$

To close this section, a modified Johnson-Cook material model is expressed for capturing the plastic response. The widely used Johnson-Cook flow stress model [48] was enhanced by Calamaz et al. [17], introducing an extra term to describe damage by tackling the strain softening issue. It includes an additional term in the flow rule σ_y that softens the material with increasing strain. In 2010, a manipulated version of their model was developed by Sima et al. [34], experimentally validated in the machining of TiAl6V4. For the ease of cross-referencing, "JC" and "MJC" are hereinafter used as the abbreviation for the terms "Johnson-Cook" and "Modified Johnson-Cook". In the metal cutting simulation of this work, the MJC model offered by Sima and Özel [34] is used. The effective yield stress value σ_y using this MJC model is obtained from

$$\sigma_y^{\mathrm{MJC}} = \underbrace{\left[A + B\left[\frac{(\bar{\epsilon}^{pl})^n}{\exp\left((\bar{\epsilon}^{pl})^a\right)}\right]\right]\left[1 + C\,ln\left[\frac{\dot{\bar{\epsilon}}^{pl}}{\dot{\bar{\epsilon}}^{pl}_0}\right]\right]\left[1 - \left[\frac{T - T_0}{T_\mathrm{m} - T_0}\right]^m\right]}_{\text{the JC terms}}\underbrace{\left[D + (1 - D)\left[\tanh\left[\frac{1}{(\bar{\epsilon}^{pl} + P)^c}\right]\right]^s\right]}_{\text{the MJC term}} \tag{18}$$

where $\bar{\epsilon}^{pl}$ is the equivalent plastic strain and the new parameters D and P are

$$D = 1 - \left(\frac{T}{T_\mathrm{m}}\right)^d \qquad P = \left(\frac{T}{T_\mathrm{m}}\right)^b \tag{19}$$

if T_0 and T_m are the reference and melting temperature, respectively. The MJC model requires the identification of five additional material parameters (*a, b, c, d, s*) in addition to the usual five parameters required for the standard JC model (*A, B, C, m, n*). The advantage is though, the original JC parameters are the same for the MJC model and do not need to be re-identified in Eq. (18), according to [34]. As shown in [34], the numerical values of these parameters can be determined by a set of SHPB tests. The values given in Table 1 are taken for the numerical simulations.

A few important remarks on the reliability of material parameters:

- It is important to emphasize that the deformation rates in SHPB tests are at least one order of magnitude lower than those occurring in cutting. This means that the material parameters for the cutting simulation must be identified from a respective cutting experiment itself. In other words, an original research on the identification of these material parameters with the help of, for instance, inverse-fitting simulations is necessitated.
- To address the previous point, Sima et al. [34] made use of a factor screening method to achieve a good agreement with the reference SHPB model. They took the data provided by Lee et al. [49] in the range of strains below 0.3 mm/mm.
- The temperature dependency of these material parameters plays a key role in their validity and accuracy, as the temperature in TiAl6V4-cutting might grow up to 1000 °K. Three different ranges of temperatures from 573 °K to 973 °K were taken in the original experiment [49], and later used by [34] to consider the impact of temperature. The values listed in Table 1 were proposed by Sima and his coworker as the best fit and are used accordingly in this paper.

3. Meshfree formulation

The formalism of the metal cutting model in a meshfree updated Lagrangian frame is revisited in this section. This matter has been exhaustively elaborated upon in [45] with a more detailed discussion. For the sake of coherence, a summary of the formulation is provided in this section. Previously mentioned in Section 1, the original adoption of SPH was for astrophysics, cosmology, and fluid dynamics applications. Due to the Lagrangian nature and their mesh-independent attribute, SPH schemes have successfully established themselves in the realm of such problems [50,51]. However, and perhaps suggested by intuition, a solid body might as well be considered as material "flow" if it undergoes severe deformations. According to Niu et al. [13], this kind of approach is often referred to as a typical "solid–flow" problem. The metal cutting process is in fact a good example of this scenario, and because of this, SPH seems to be a prime candidate to investigate it. That explains why skipping the daunting mesh re-computation task needed for mesh-dependent analyses has been frequently reported as a salient power of SPH methods in solving highly dynamic solid problems. That being said, the governing equations presented in the preceding section need to be first discretized and then coupled with a proper flow–stress constitutive model given in Eq. (18).

3.1. Spatial discretization

In the following, the ingredients required for discretizing the governing equations in Section 2.2 are reviewed. This chapter is not aimed at providing a full derivation procedure of SPH, but rather outlining the important expressions necessary for a clearer understanding of the

Table 1
Parameters used for the MJC model of TiAl6V4, in accordance with *model 3* in [34].

	A	B	C	a	b	c	d	m	n	s
Unit	[MPa]	[MPa]	[–]	[–]	[–]	[–]	[–]	[–]	[–]	[–]
Value	724.7	683.1	0.035	2.0	5.0	2.0	1.0	1.0	0.47	0.05

proposed solver. More detailed and complete discussions on meshfree methods can be found in the standard text books and technical reports, e.g., [51–53].

3.1.1. Function & kernel approximation

A given field function f at point i in 2D space is represented by $f(\underline{x}_i) = f(x_i, y_i)$ and its first derivative are approximated with SPH. Using any meshfree method including SPH, a finite number of particles is used to discretize the body for approximation purposes. In essence, a numerical interpolation is carried out on these particles through a smoothed weighted averaging procedure. The weight of each integration point is defined by applying a kernel function. In analogy with Li and Liu [53], the integral form of f using SPH approximation at off-sample location(s) $\underline{\tilde{x}}$ is recalled

$$\left\langle f(\underline{\tilde{x}})\right\rangle^{\mathrm{SPH}} = \int_{-\infty}^{+\infty} f(\underline{x}')W_h(\underline{\tilde{x}} - \underline{x}', h)dx' \tag{20}$$

where a smoothing kernel function W_h has the smoothing length h that controls the amount of mollification (i.e., smoothing) in this kernel. For more legibility, the term $W_h(\underline{x}_i - \underline{x}_j, h)$ is hereinafter replaced with the shortened form W_{ij}. The SPH approximation in Eq. (20) was initially derived by Gingold and Monaghan [1], Lucy [2] and can be integrated using a Riemann sum (or the trapezoidal rule) over a set of N points, resulting in

$$\left\langle f(\underline{x}_i)\right\rangle^{\mathrm{SPH}} \approx \sum_{j=1}^{N} V_j\, f(\underline{x}_j)W_{ij} \tag{21}$$

if $V_j = m_j/\rho_j$ is the integration weight of particle j. The smoothing kernel function, W_{ij}, is the central element in all meshfree methods from both stability and accuracy points of view. A proper kernel should be continuous, positive definite in its support, and at least as many times differentiable as the degree of the derivative to be approximated. The latter means that the kernel has to be at least two times differentiable for approximating second derivatives. In favor of computational efficiency, a cut-off radius is often introduced as the compact support to the kernel function. By doing so, the neighboring search procedure is restricted to a much smaller limited range. The widely-used “Cubic B–spline” kernel function is employed

$$W_{ij} = n_c \times \begin{cases} 1 - \frac{3}{2}q^2 + \frac{3}{4}q^3, & 0 \le q < 1 \\ \frac{1}{4}(2-q)^3, & 1 \le q \le 2 \\ 0, & \text{otherwise} \end{cases} \tag{22}$$

with the normalization factor $n_c = 10/7\pi h^2$ for 2D space given by [54,55], and $q = |\underline{x}_{ij}|/h$.

3.1.2. First derivative approximation

Calculating the first derivatives of $f(\underline{x}_i)$ is performed by applying the gradient operator on the kernel function in Eq. (21). As it was also utilized by [51,52], one can alternatively derive anti-symmetric and symmetric forms of SPH via

$$\left\langle \nabla f(\underline{x}_i)\right\rangle^{\mathrm{SPH\text{-}asym}} = \sum_{j=1}^{N} V_j \left(f(\underline{x}_j) - f(\underline{x}_i)\right)\nabla W_{ij} \tag{23}$$

$$\left\langle \nabla f(\underline{x}_i)\right\rangle^{\mathrm{SPH\text{-}sym}} = \rho_i \sum_{j=1}^{N} m_j \left(\frac{f(\underline{x}_i)}{\rho_i^2} + \frac{f(\underline{x}_j)}{\rho_j^2}\right)\nabla W_{ij} \tag{24}$$

$$\nabla W_{ij} = \frac{\underline{x}_{ij}}{|\underline{x}_{ij}|}\frac{\partial}{\partial q}W_{ij} \tag{25}$$

The SPH approximations presented in Eqs. (23) and (24) are employed for replacing the velocity and pressure gradients in Eq. (1) and Eq. (2). The discussion with respect to second derivative, and subsequently the temperature gradient, follows.

3.1.3. Second derivative approximation

The second derivatives only appear in the Laplacian form in the conduction part of Eq. (3). Since the SPH approximation of higher order derivatives is prone to instability and even less accurate [56,57], it is of utmost importance to adopt a robust scheme for solving the heat equation. Thus, a Particle Strength Exchange (PSE) type kernel is used in this work, motivated by a very appealing accuracy vs. efficiency tradeoff that it offers. Considering the PSE approach, a target kernel, W^{PSE}, is derived for the desired differential operator. W^{PSE} is imposed to deliver the favorable order of accuracy (here 2). Following the PSE methodology [56,58–60], the Laplacian of function $f(\underline{x}_i)$ in 2D space is computed from

$$\left\langle \nabla^2 f(\underline{x}_i)\right\rangle^{\mathrm{PSE}} = \frac{1}{h^2}\sum_{j=1}^{N} V_j \left(f(\underline{x}_j) - f(\underline{x}_i)\right)W_{ij}^{\mathrm{PSE}} \tag{26}$$

$$W_{ij}^{\mathrm{PSE}} = \frac{4}{\pi h^2}\exp(-|q|^2) \tag{27}$$

Comparing Eq. (26) with Eq. (23), the Laplacian approximation is not found by performing the gradient operator on W twice, but devising an *ad hoc* kernel W^{PSE} for this differential operator ∇^2. A list of various PSE kernels up to 2D applications is given by Eldredge et al. [56]. Interestingly enough, no further consideration is required with respect to the thermal boundaries in this work, since the PSE scheme is inherently capable of satisfying the adiabatic condition.

3.1.4. Corrected kernels

Due to the boundary deficiency, the conventional SPH method presented in Eq. (21) and (23) is unable to reconstruct even a linear function in the entire domain [51,61]. This well-known drawback in meshfree simulations would be even more problematic if a first-order consistent (i.e., linear-complete) approximation of the spatial derivatives is entailed by the algorithm used. Among several existing remedies to this shortcoming, the Corrected Smoothed Particle Method (CSPM) is employed for a comparison purpose in this work. Initially derived by [62], the corrected scheme can be written in compact form as

$$\left\langle \nabla f(\underline{x}_i)\right\rangle^{\mathrm{CSPM}} = \underline{\underline{A}}_i^{-1}\left[\sum_{j=1}^{N} V_j \left(f(\underline{x}_j) - f(\underline{x}_i)\right)\nabla W_{ij}\right] \tag{28}$$

$$\underline{\underline{A}}_i = \sum_{j=1}^{N} V_j\left(\underline{x}_{ji} \otimes \nabla W_{ij}\right) \tag{29}$$

in which $\underline{\underline{A}}_i$ is the second-order CSPM re-normalization tensor for first derivatives at particle i, and $\underline{x}_{ji} = \underline{x}_j - \underline{x}_i$.

3.2. Discretized governing equations

Taking into account the standard conservation laws provided in Eqs. (1)–(3), the system of governing PDEs is consequently approximated by means of the SPH method just presented. If the gradient approximations formulated in Eqs. (23) and (24) are applied to Eqs. (1) and (2), and the Laplacian approximation presented in Eq. (26) is applied to Eq. (3), the system of conservation equations arrives at

$$\left\langle \dot{\rho}\right\rangle_i = -\rho_i \sum_{j=1}^{N} V_j\,(\underline{v}_j - \underline{v}_i)\cdot\nabla W_{ij} \tag{30}$$

$$\left\langle \dot{\underline{v}}\right\rangle_i = \sum_{j=1}^{N} m_j \left(\frac{\underline{\underline{\sigma}}_i}{\rho_i^2} + \frac{\underline{\underline{\sigma}}_j}{\rho_j^2} + \underbrace{\Pi_{ij}\underline{\underline{I}} + \underline{\underline{R}}_{ij}}_{\text{stabilizers}}\right)\cdot\nabla W_{ij} + \frac{1}{m_i}\underline{b}_i \tag{31}$$

$$\left\langle \dot{T}\right\rangle_i = \chi\left(\frac{\sigma_y\,\dot{\varepsilon}^{pl}}{\rho c_p}\right)_i + \frac{a}{h^2}\sum_{j=1}^{N} V_j\,(T_j - T_i)W_{ij}^{\mathrm{PSE}} \tag{32}$$

$$\left\langle \dot{\underline{x}}\right\rangle_i = \underline{v}_i + \underbrace{\beta\sum_{j=1}^{N}\frac{m_j}{\rho_i + \rho_j}(\underline{v}_j - \underline{v}_i)W_{ij}}_{\text{XSPH correction}} \tag{33}$$

with the thermal diffusivity denoted by $a = k/\rho c_p$ in Eq. (32). Moreover, the stabilization terms in Eq. (31) as well as the XSPH correction in Eq. (33) are incorporated. A summary of these additional terms follows.

- “Π_{ij}**– Artificial Viscosity”** It has been shown in many applications such as [51,61], that the incidence of oscillatory modes, particularly in shock wave problems, can be suppressed by the use of a viscosity type stabilizer. As a matter of fact, this was first introduced for SPH shock simulations by Monaghan and Gingold [63]. The artificial viscosity term Π_{ij} can alleviate the oscillatory modes issue with dissipating the energy produced by perturbations. Although the introduction of this stabilization term brings out some new non-physical parameters, the improvement it makes is unquestionable. This is also a quite useful remedy to prevent from the numerical divergence when two approaching particles get too close [64]. Successfully employed for an orthogonal cutting test with SPH, the numerical values of the stabilization factors in this work are chosen in accordance with [13].
- “$\underline{\underline{R}}_{ij}$**– Artificial Stress”** In order for the linear momentum equation to guard against the tensile instability, an artificial stress tensor $\underline{\underline{R}}_{ij}$ is also added to Eq. (31). This idea was first used by Gray and Monaghan in the context of elasto-dynamics [65], and later exploited for CFD applications [66]. Roethlin et al. [14] give a good elaboration on this subject, demonstrating the functionality of this approach in a classical rubber ring impact test. The same tuning parameters as Röthlin et al. [14] are herein used for all simulations.
- **“XSPH Correction”** Suggested by Monaghan [67], an effective manipulation can be applied to the advection field in order to hinder the particles ensemble from severe clustering or dispersing situations. Utilizing the XSPH approach, particle’s position is updated by a modified velocity field. As expressed in Eq. (33), the XSPH modified velocity field consists of the current momentum velocity obtained from Eq. (31) plus the average of particles within the smoothing kernel around the source particle. For the numerical simulations in this work $\beta = 0.5$ is used.

3.3. Temporal discretization

After the spatial discretization, a set of ordinary differential equations exists, which needs to be discretized in time. That is, the state variables and displacements presented in Eqs. (30)–(33) are evolved over time. To do so, a second-order explicit leapfrog integration scheme adopted for the SPH implementation by Monaghan [67] is utilized. It constructs the predictor and corrector steps by updating positions and velocities at interleaved time points [68,69]. This choice is made due to the fact that the leapfrog scheme is in fact a second-order accurate method in time, while it requires only one evaluation of functions per step (similar to the first-order Euler scheme). The maximum time-step size is initially calculated as the smaller of the maximum step size of the mechanical and thermal parts. That is, a CFL-type condition with a CFL constant within the range of [0.15,0.35] is determined for each of the two parts (i.e., mechanical and thermal), and stays unchanged during the simulation. For the multi-resolution tests in this work, the step size associated with the finer zone governs the entire solution stability, and is taken accordingly.

4. Spatial Refinement in Meshfree Methods

In this section, an overview of the spatial refinement strategy is presented. The algorithm is merely based upon particle splitting, has been already used for a broad array of other applications [18,19,24], and is straightforward. Given the refinement pattern centered about the original particle position, candidate “parent” particles are split into several “child” particles. Four refinement patterns are employed in this work, depicted schematically in Fig. 3. The goal is to increase the resolution efficiently, and only in certain regions of the domain. The refinement approach presented in this section follows the procedure elaborated upon

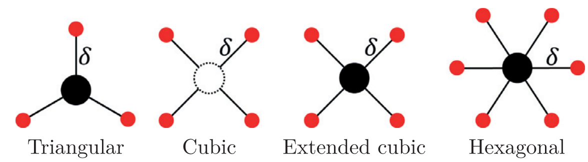

Fig. 3. 2D refinement patterns used in this work.

by Feldman and Bonet [24]. One refinement step per particle is considered for the present results, while the implementation allows for performing multi-level refinement in the code.

In order to implement dynamic refinement into a working code a number of considerations need to be dealt with. The whole refinement stages are executed in a separate unit of the code, henceforth referred to as “refinement block”. Firstly, the workability of the solver should be independent of the refinement block. Secondly, developing a general algorithm whereby the newly added particles ensure the conservation of the basic properties of the entire system, which is a major concern. The step-by-step procedure described in Algorithm 1 are the contents of refinement block included in Fig. 5.

Algorithm 1 Spatial refinement procedure via splitting of a parent particle.

1: **Identification:** given the criterion, candidate particles are marked as “parent”.
2: **Action:** given the pattern, the “parent” particles are replaced with their “child” particles.
3: **Mass Splitting:** divide the mass of each parent uniformly into its child particles.
4: **Setting** h**:** set the smoothing length of each child according to a prescribed coefficient.
5: **Primary Assignment:** assign the velocity and density of parent into its child particles.
6: **Secondary Assignment:** if necessary, use Eq. (21) for the SPH interpolation.

The aforementioned stages of the refinement block should be applied automatically by the proposed code at runtime with no intervention required. This matter is illustrated with the help of a flowchart given in Fig. 5.

4.1. Refinement criteria

Candidate particles for refinement could be identified by several user-defined criteria depending on the type of the problem at hand. For the metal cutting process, the following classification seems appropriate:

- Fixed refinement zone(s)
- Moving refinement frame(s)
- Number of neighbors (i.e., the total number of neighbors located inside the support domain)
- Problem-dependent criteria:
  - Critical absolute value (e.g., temperature, plastic strain, von Mises stress, etc.)
  - Critical gradient/divergence value (e.g., divergence of velocity, strain gradient, etc.)

The aim of this paper is to implement a refinement algorithm that is versatile and independent of the criterion used. In the numerical simulations of this paper, only the first 2 criteria are exploited. A concise argumentation to support this selection follows.

As was discussed in [32] as well, multi-resolution and adaptive schemes can make use of various criteria suited for the problem at hand. Notwithstanding, opting for a single value threshold or the gradient thereof prove to be a reasonably good choice in most cases. Selecting an adept threshold value is yet another concern, which needs to be linked with the associated application. As far as metal cutting is concerned, a quite useful choice by which the sensitive areas (i.e., shear zones and vicinity of cutting edge of the tool) are always covered, can be a moving refinement frame. The geometry of this frame is sensitive to the settings of the cut: the cutting speed, the target feed, the rake and clearance angles. While not being the most efficient option in terms of the computational cost, such a criterion is nevertheless used for the metal cutting simulation in this work. Comparing the stability and accuracy issues of other criteria in this application requires an original research, and will be an outlook of the present paper.

4.2. Refinement patterns

In general, the total number of unknown variables for each child particle in 2D applications is six. This includes the mass, position, velocity, and smoothing length. These six unknowns need to be determined for each newly born child particle (please mind the vector notation).

$$m_{\text{child}} = ? \qquad \underline{x}_{\text{child}} = ? \qquad \underline{v}_{\text{child}} = ? \qquad h_{\text{child}} = ? \tag{34}$$

Inspired by a variety of successful exploitations in the literature (e.g., [24,26,32]), four different patterns are examined. As displayed in Fig. 3 through left to right, these four templates are defined in a way that the parent particle is represented by solid black, whereas its corresponding child particles are shown in solid red. In all configurations except the cubic one, the parent particle remains active after splitting, and will be located in the center of the stencil, at the position of the original parent particle. In other words, there will be a child at the exact place of the former parent.

In [26,30], it was claimed that the hexagonal pattern reveals a good balance between the total number of particles and the reduction of the density error. Nevertheless, any judgment regarding the optimal arrangement is application dependent, and can be made only if a comprehensive error analysis is performed on the alternative patterns. Since it is not planned in this study to solve a global minimization problem for exploring the most accurate arrangement, the only unknown variables in each pattern will be the dispersion coefficient κ and smoothing ratio ξ. These two characteristic scalar parameters κ and ξ are defined. The dispersion value, δ, is a coefficient of the initial particle spacing as $\delta = \kappa \Delta x$, to define the distance between the child particles and their parent (as shown in Fig. 3). A useful rule of thumb is to take the condition $0.2\Delta x < \delta < 0.4\Delta x$ into account depending on the splitting pattern used. Afterwards, the smoothing ratio ξ needs to be set for calculating the smoothing length of the child particles via $h_{\text{child}} = \xi h_{\text{parent}}$. These two parameters do control the spatial coherence of the refined zone and the smoothing ratio of the smaller particles.

Given the refinement pattern, the total number of six unknowns for each child particle can be reduced immediately by specifying the child locations. That is, each individual pattern follows a pre-specified configuration that defines both the number and relative locations of the child particles. Unless otherwise mentioned in this manuscript, it is assumed that the mass of parent particle is equally split into its child particles. This means, the mass and position of each child particle have already been determined and it is left to find the other two unknowns in (34), i.e., the velocity and the smoothing length.

Fig. 4 shows the situation before and after splitting. In this figure, the red (parent) particle meets the chosen refinement criterion and is colored red. It is then split into 5 child particles by applying the extended cubic pattern, leading to a refined configuration.

After refinement, the smoothing length of a child particle is different from the adjacent particles which have not been split. While we realize that the optimal choice in situations where there are areas of widely differing smoothing lengths would be a k-d tree by [70] (an adaptive space-partitioning data structure for organizing points), our implementation still relies on a simple spatial hashing algorithm and the cell-list data structure [71], even with refined particle simulations. This means

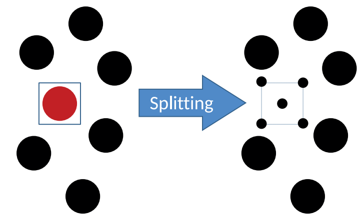

Fig. 4. Schematic of particle splitting. Particles with lower mass are depicted with smaller size.

that we simply choose twice the maximum smoothing length in the simulation as the edge length of the spatial hashing structure, which entails that refined particles will most likely check more neighbors as potential interaction partners than necessary.

4.3. Refinement procedure

Pragmatically speaking, it is not possible to conduct the splitting procedure without introducing any errors. What seems necessary, though, is to make sure that the introduced errors are minimum or negligible for the desired application. Without solving a non-linear minimization problem for finding the optimal mass distribution of the child particles, a uniform mass distribution approach is adopted here which splits the mass proportional to the refinement pattern employed (see Fig. 3 together with Table 2). This trivial approach ensures the mass conservation of the system in the simplest way. It can be perceived from the numerical results of the metal cutting simulation in Section 6 that the equal mass approach works very well, and is hence a good choice for this case.

Also, the linear momentum conservation automatically holds true by enforcing that each child particle has the same velocity and density of the corresponding parent particle. There exists a variety of other approaches concerning this issue, mostly utilized in the area of CFD applications. For example, Vacondio et al. [30] proposed a preset constant value for the refinement parameters κ and ξ, and then obtaining the masses of the “child” particles by minimizing the global error between the refined and unrefined density. Their method follows the same global error-minimization methodology originally developed by Feldman and Bonet [24] for SPH fluid flow problems. A more thorough discussion on these conservation issues can be found in [26,28,32].

Table 2 summarizes the magnitudes of the corresponding quantities for each pattern wherein δ represents the dispersion value. These quantities include the number of child, their masses, and coordinates of their relative positions. For more readability, the subscripts “c” and “p” are used instead of “child” and “parent”.

A more accurate and consistent approach would probably include an *a posteriori* correction stage, whereby the density of the child particles are re-normalized and their velocities are corrected via higher order accurate schemes. This consideration was taken into account for fluid flow applications by Hu et al. [28] and is not implemented in the present work. This section is followed by a comparison benchmark, which is carried out in Section 5. Upon this estimation, an efficient pattern will be picked for the final metal cutting simulation. To sum up, the logic of the present solver is illustrated with the help of a flowchart given in Fig. 5.

5. Preliminary study

In this model example, the density refinement errors are calculated as a preliminary test on 4 alternative configurations. Upon this assessment, the optimal refinement pattern is explored to be used for the metal cutting simulation in Section 6.

The density is approximated in a unit square discretized by 11 particles in each direction, resulting in a total number of 121 unrefined particles. The density is assumed to be $\rho = 1$ in the whole domain. Consistent

Table 2
Summary of the parameters used in the present refinement approach.

Pattern	#child	m_c	x_c	y_c
Triangular	3	$m_p/4$	$\{x_p, x_p \pm \cos(\frac{\pi}{6})\delta\}$	$\{y_p, y_p + \delta, y_p \pm \sin(\frac{\pi}{6})\delta\}$
Cubic	4	$m_p/4$	$\{x_p \pm \cos(\frac{\pi}{4})\delta\}$	$\{y_p \pm \sin(\frac{\pi}{4})\delta\}$
Extended cubic	4	$m_p/5$	$\{x_p, x_p \pm \cos(\frac{\pi}{4})\delta\}$	$\{y_p, y_p \pm \sin(\frac{\pi}{4})\delta\}$
Hexagonal	6	$m_p/7$	$\{x_p, x_p \pm \delta, x_p \pm \cos(\frac{\pi}{3})\delta\}$	$\{y_p, y_p \pm \delta, y_p \pm \sin(\frac{\pi}{3})\delta\}$

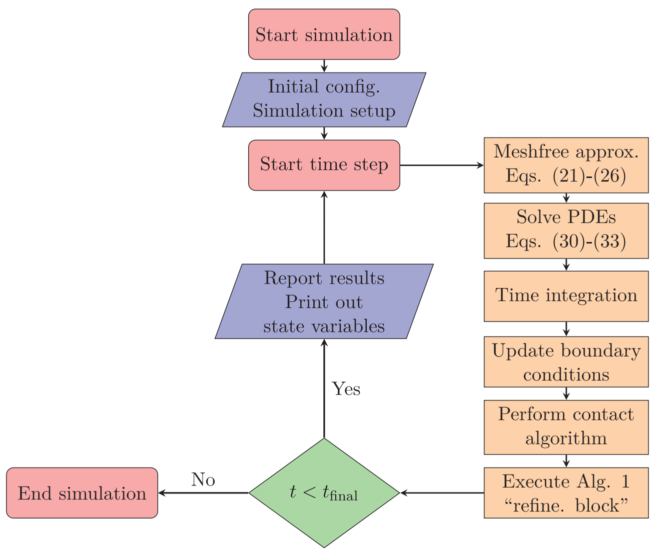

Fig. 5. Flowchart of the model logic for each time-step.

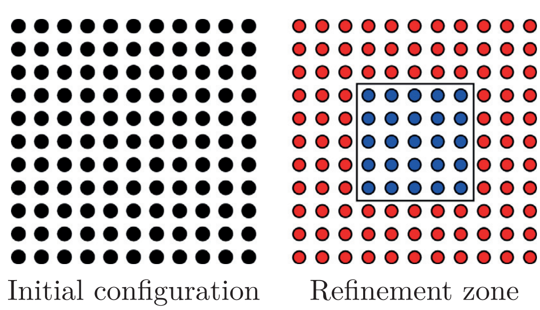

Fig. 6. Initial configuration of the preliminary example. Blue particles in the right figure are to be refined.

with the initial configuration in the model used by Feldman [72], the same refinement zone is herein considered, as shown in Fig. 6. The central 5×5 region of particles is then refined using the four selected patterns. In all cases, the mass of parent particle is uniformly split amongst its child particles. Depicted in Fig. 7 is the spatial arrangement of particles at each pattern after the splitting procedure. In a mathematically non-rigorous fashion, the certain (κ, ξ) pairs chosen for each pattern are identified as good choices that result in sufficiently small refinement errors. The magnitudes of the refinement parameters (κ, ξ) used are given at the end of this section, in Table 3.

Table 3
Numerical values of the error norms.

Pattern	$\mathrm{N}_{\mathrm{total}}$	κ	ξ	$\mathcal{L}_\infty$	$\mathcal{L}_2$
Triangular	196	0.4	0.6	0.2730	0.0868
Cubic	196	0.35	0.6	0.1445	0.0588
Extended cubic	221	0.3	0.6	0.1742	0.0595
Hexagonal	271	0.4	0.6	0.1995	0.0675

In the next step, the SPH function approximation in Eq. (21) is carried out by employing the cubic B–spline kernel given in Eq. (22). The smoothing length of the unrefined configuration is chosen to be $h = \Delta x$ for all patterns. Finally, the density approximation error is calculated by comparing the particles density before and after refinement and computing its absolute relative value as $err = |\rho^{\text{before}} - \rho^{\text{after}}|/\rho^{\text{before}}$ for all configurations. Fig. 8 demonstrates the distribution of this error in the computational domain. On the top row, the colorbars are variable with the respective local maxima for the four patterns. On the bottom row, a fixed colorbar is set with the global maximum value to provide a better comparison. As expected, the discrepancy is larger in the regions where the refined and unrefined particles meet.

In order to present a quantitative comparison as well, the following error norms are computed in discrete form for particle i inside the computational domain Ω

$$\mathcal{L}_\infty = \max_{i\in\Omega} |\rho_i^{\text{after}} - \rho_i^{\text{before}}| \tag{35}$$

$$\mathcal{L}_2 = \sqrt{\frac{1}{\mathrm{N}_{\text{total}}} \sum_{i=1}^{\mathrm{N}_{\text{total}}} \left(\rho_i^{\text{after}} - \rho_i^{\text{before}}\right)^2} \tag{36}$$

if $\mathrm{N}_{\text{total}}$ indicates the total number of particles. $\mathcal{L}_\infty$ shall designate a measure of the maximum local error and $\mathcal{L}_2$ the average global measure of error over the entire domain. It is important to explain how the calculation in Eq. (35) and Eq. (36) is carried out. The density of particles outside the refinement zone shown in Fig. 6 is known before and after refinement. For the locations inside the refinement zone, the initial value $\rho = 1$ is assigned to the density of child particles before refinement and compared with their respective values after refinement. Please note that $\mathrm{N}_{\text{total}}$ in this example is associated with the configuration after refinement (i.e., the sum of red and blue particles in Fig. 7). The respective

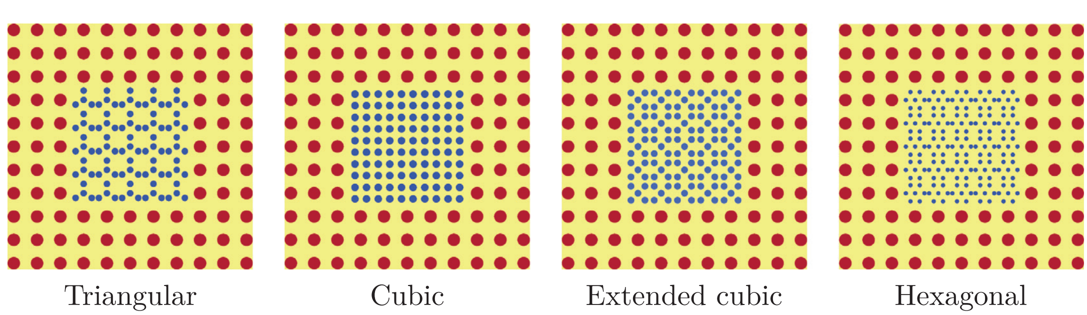

Fig. 7. Particles arrangement in the 4 patterns used. Mass of particles is color-coded: low (blue) & high (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

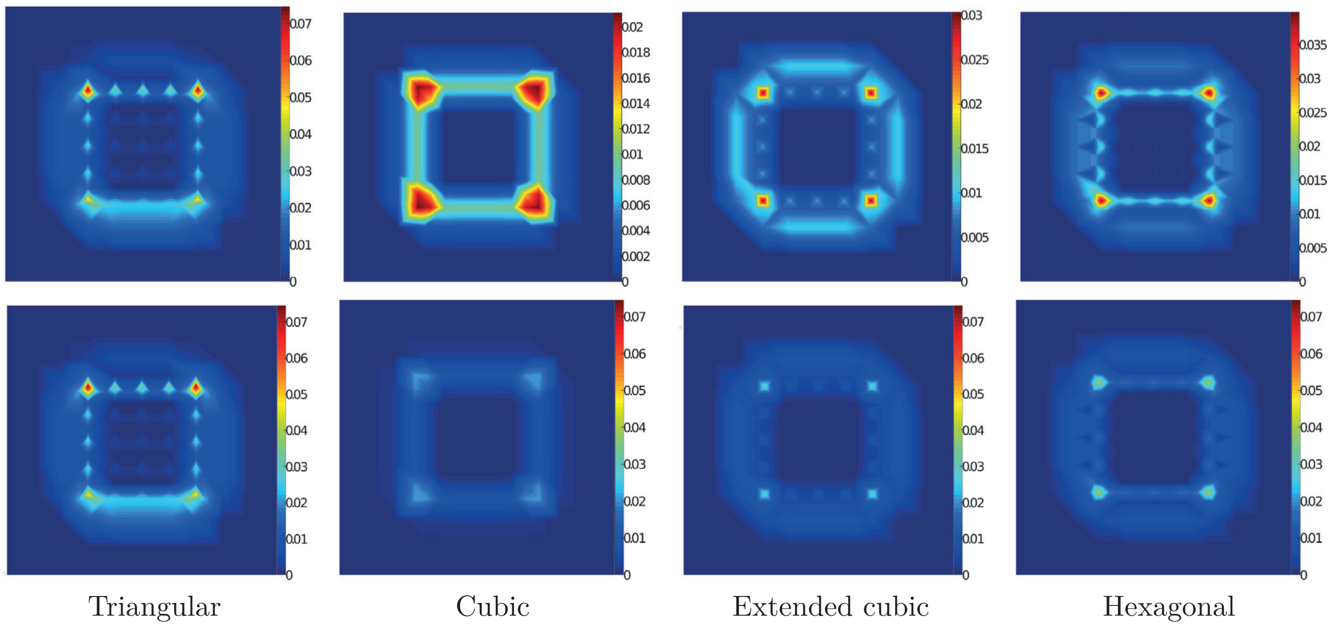

Fig. 8. Global density refinement errors using 4 different patterns. Top row: colorbar with local maxima; bottom row: colorbar with global maximum. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

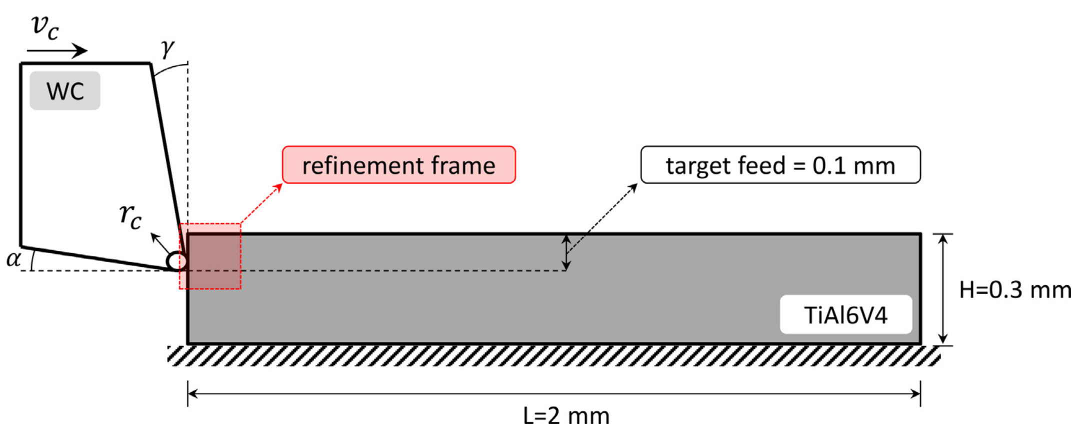

Fig. 9. Sketch of the cutting geometry and the refinement frame, with the parameters given in Table 4. The cutting tool is a rigid body and the workpiece is made of TiAl6V4. The dimensions are sufficiently large such that the solution is not affected by boundary conditions.

errors for the four patterns shown in Fig. 3 are summarized in Table 3 together with the values chosen for κ and ξ.

Having performed an error analysis, the "cubic" pattern is picked for the metal cutting simulation. This conclusion is drawn upon the lower error norms, but also adding a lower number of child particles compared to the other arrangements (see Table 3). Meanwhile, the cubic pattern is able to provide a more homogeneous distribution in the refined configuration, as can be seen in Fig. 7 as well.

6. Metal cutting simulation

This section revolves around the performance investigation of the solver presented, applied to a 2D metal cutting problem. The case study pertains to a rigid tool, cutting a workpiece clamped from the bottom surface. Drawn in Fig. 9 is the sketch of the considered metal cutting geometry, followed by the associated parameters listed in Table 4. The same initial configuration as [34] is taken in order to assess the reproducibility of the proposed meshfree solver for the verification purposes. The target length of cut is l = 1 mm, 50% of the total length of the workpiece. This cutting length entails $\Delta t = 0.12$ ms of the processing time, given a cutting speed $v_c = 500$ m/min used for the qualitative purpose.

Table 4
Geometric parameters used for the cutting test.

Body	Property	Symbol	Unit	Value
Tool	Clearance angle	α	deg	11.0
	Rake angle	γ	deg	0.0
	Cutting edge radius	r_c	mm	0.005
	Speed	v_c	m min^{-1}	{120, 241, 500}
Workpiece	Length	L	mm	2.0
	Height	H	mm	0.3
	Depth of cut	d	mm	0.1

It is worthwhile to describe some implementation aspects of this test problem in the results of SPH. Fig. 10 gives an illustration of the fixed boundary condition at the bottom of the workpiece. As can be seen, both velocity $\underline{v}$ and acceleration $\underline{a}$ of all (red) boundary particles are set to zero at each time-step. By doing so, the respective solid boundary condition is ensured during the simulation. In Fig. 11, the moving refinement frame is shown and shaded in red, as the refinement criterion used for this application. It clarifies how the big (unrefined) particles are split into smaller (refined) particles using the cubic pattern as soon as they are found inside the bounding box of this refinement frame.

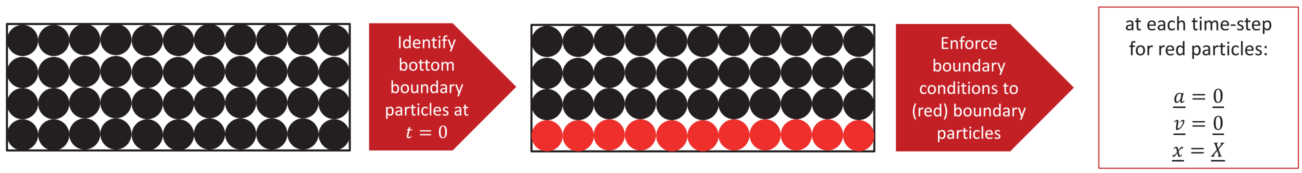

Fig. 10. Schematic illustration of how the fixed boundary condition can be applied to SPH particles at the bottom of the TiAl6V4 workpiece. The symbols $\underline{X}$ and $\underline{x}$ in the rightmost box denote the position of each (red) boundary particle on the original and current configuration, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

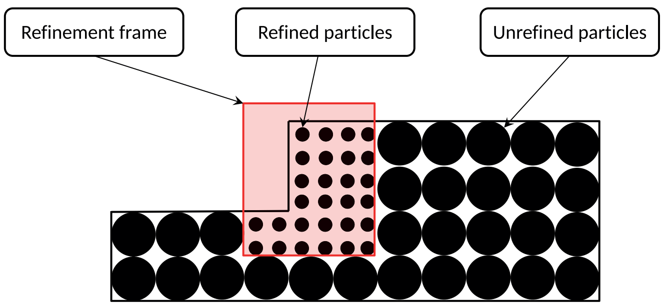

Fig. 11. Graphical demonstration to show how the refinement scheme described in Section 4 works for a simplified cutting test.

Table 5
Physical properties used for the cutting test.

Body	Property	Symbol	Unit	Value
Tool	Density	ρ	kg m^{-3}	14700
Workpiece	Density	ρ	kg m^{-3}	4430
	Young's modulus	E	GPa	113.8
	Poisson ratio	ν	–	0.35
	Heat conductivity	k	W m^{-1} K^{-1}	7.3
	Specific heat capacity	c_p	J kg^{-1} K^{-1}	580.0
	Reference temperature	T_0	K	298
	Melting temperature	T_m	K	1878
	Taylor-Quinney factor	χ	–	0.90
Tool-Workpiece	Coefficient of friction	μ	–	0.35

All simulations are carried out without adapting the smoothing length, where a constant $h = 1.5\Delta x$ is always used. Clearly, the initial particle spacing Δx for particle i in multi-resolution cases must be assigned considering the zone that particle i belongs to. Unlike [34], the mechanical as well as thermo-physical properties of TiAl6V4 given in Table 5 are constant and temperature independent. The Coulomb friction model with a constant coefficient is utilized.

The following four models are examined, through which the respective improvement gained by the refinement procedure will be explored. The parameter N_0 indicates the total number of particles in the beginning of the simulation.

- *Model 1*: Single low resolution ($N_0 = 6231$)
- *Model 2*: Multi resolution with dynamic refinement ($N_0 = 6807$)
- *Model 3*: Multi resolution with *a priori* refinement ($N_0 = 15847$)
- *Model 4*: Single high resolution ($N_0 = 24461$)

Based on the preceding test in Section 5, the cubic pattern with uniform division of the mass is selected for *Model 2* and *Model 3*. In *Model 3*, only the upper half of the workpiece is refined *a priori* in the initial configuration setup. In *Model 2*, the moving refinement frame is attached to the tool tip and is moved with the cutting speed. This frame has a width of $l_x = 0.25$ mm and height of $l_y = 0.15$ mm initially. The frame size can be dynamically adjusted during the simulation, if necessary. Only one refinement step is permitted per particle, and particles are not allowed to return to their original resolution, even if they leave the refinement frame. The dynamic refinement strategy employed in the present work is deliberately kept as simple as possible such that the follow-up step towards the optimal adaptivity in metal cutting simulations can be pursued with minimum effort.

Aligned with the flowchart presented in Fig. 5, the simulation proceeds by solving Eqs. (30)–(33) linked to the yield stress in Eq. (18). Already discussed in Section 3.2, the artificial stress and viscosity are added to the momentum equation. The advection field is also regulated via the XSPH correction term. The heat transfer from the workpiece to the tool is not taken into account, and the heat conduction is solved in the workpiece with adiabatic boundaries. The performance of the present in-house code is herein assessed with results of two kinds, but have also been evaluated in numerous examples computed satisfactorily before [9,10,14,60,73,74]:

1. Quantitative investigation of the cutting forces and chip shape
2. Qualitative investigation of the selected state variables (equivalent plastic strain and temperature)

In the end, the associated computational effort of each model is summarized, suggesting the efficiency of the dynamic refinement implemented in *model 2*.

6.1. Quantitative assessment

At first, predicted forces from the meshfree simulations are compared with the measured forces reported in [34]. The same cutting geometry and material data (see Table 1) offered by Sima and Özel [34] are used for reproduction. Given the same cutting speed $v_c = 120$ m/min as [34], the simulation is run until the processing time $\Delta t = 0.25$ ms. A schematic diagram is given in Fig. 12, illustrating how a query particle p is first permitted to enter the bounding box of the tool, and then pushed out by applying a penalty contact force proportional to its (virtual) penetration depth (parameter d in Fig. 12). A Free-Body-Diagram (FBD) of this particle is also plotted in this figure, including the contact and friction forces (which are perpendicular to each other), color coded in blue and green, respectively. The resultant cutting (F_c) and normal/thrust (F_t) forces of the tool are measured by summing the contributions of all contacting particles.

Fig. 13 depicts the evolution of fores in time for the 4 meshfree models used, where the black dashed lines are the experimental measurements by Sima and Özel [34]. Table 6 conveys that the average cutting force using the meshfree method is in agreement, whereas the average normal force (a.k.a. thrust/passive forces) is not. The normal forces are underestimated with an average error of about 50%. One may argue that underestimation of normal forces is not necessarily inherent to meshfree methods only. In fact, it is known as a general trend in metal cutting simulations which happens also in FEM analyses, e.g. the results published by Jawahir et al. [75].

A justification for this noticeable discrepancy could indicate the importance of the tool wear modeling. It was also discussed by Niu et al. [13] that the rigid body assumption for the cutting tool results in underprediction of the normal forces. Besides, the tool-chip contact length and frictional behavior can directly influence the normal force values. More sophisticated material models and/or numerical treatments in this regard could possibly improve the accuracy of the results. All in one, the computed values shown in Fig. 13 are fairly acceptable, and shall approve the credibility of the proposed method.

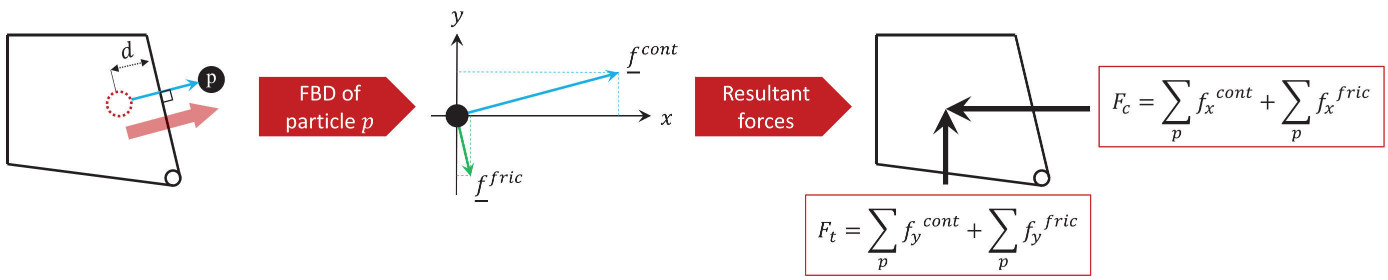

Fig. 12. Schematic diagram of how the cutting and normal forces are computed for the tool. First, the contact algorithm is established for a query particle p as soon as it is found inside the bounding box of the tool. Since this location in not admissible, particle p is pushed out by exerting a penalty contact force onto it, proportional to its (virtual) penetration depth d. The x and y components of both contact and friction forces are summed for all particles found inside the tool, leading to the resultant F_c and F_t magnitudes of the cutting tool.

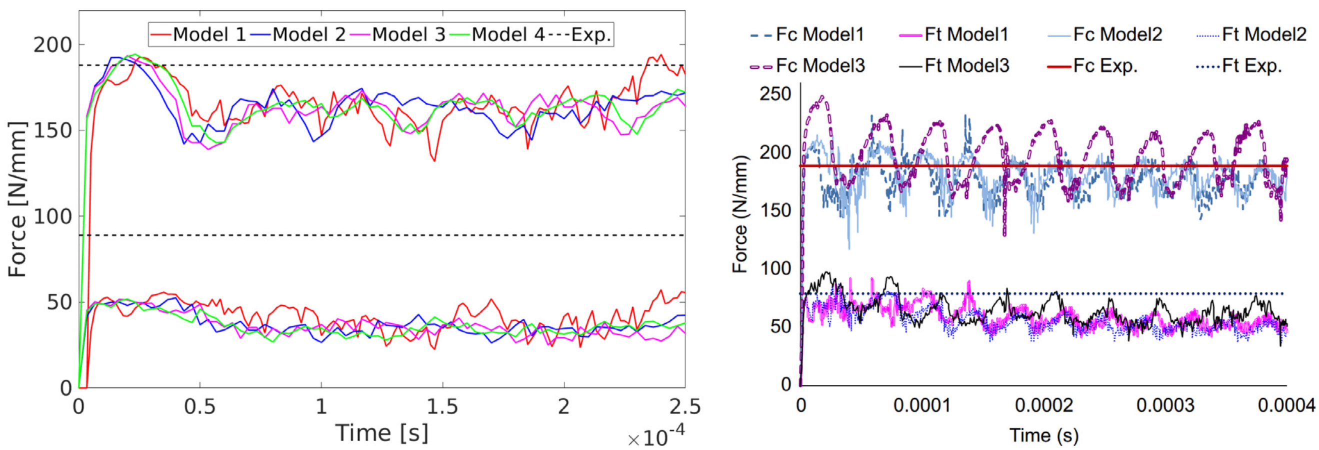

Fig. 13. Comparison of forces in time: meshfree results (left) and FEM vs. experimental measurements by [34] (right). The cutting and thrust (or normal) forces are denoted by F_c and F_t in the FEM reference data, respectively.

Table 6
Summary of the average forces using different methods.

Component	Meshfree	FEM [34]	Experiment [34]
Cutting force [N/mm]	167	195	189
Normal force [N/mm]	39	62	90

Table 7
Values of serrated chip segments in Fig. 15.

#Segment	1	2	3	4	5	Average
Length [µm]	83	88	63	68	83	**77**

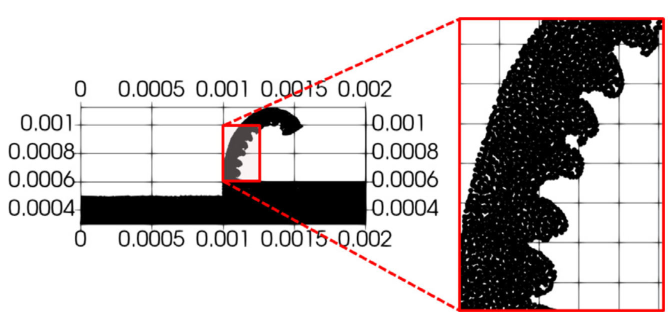

Fig. 14. Close-up of the chip segments in *Model 4*.

Table 8
Comparison of (average) simulated serrated chip with the respective data from [34].

Component	Meshfree	FEM [34]	Experiment [34]
Length [µm]	77	35	80

Afterwards, the distance between the chip segments is chosen as another measure of quantitative comparison. The reference experimental data is picked from [34] produced under the same cutting geometry with $v_c = 241$ m/min. Unless a separate geometrical tool is utilized for capturing the interface in a particle-based solver, any judgment on the geometric features is most likely linked to the resolution used. An approximate calculation of the pixels in the magnified frame of Fig. 14 is carried out, which turns out to be a good indicator for this cutting application. Illustrated in Fig. 14 is a close-up of the serrated chip pitch for 6 successive teeth, followed by the comparison of the predicted chip shape and the reference image in Fig. 15. The calculated values for the segments 1–5 in Table 7 are coupled with the corresponding segment number in Fig. 15.

Yielding an average error of roughly 4% for the segments in Fig. 15, an excellent agreement with experimental data of the same condition can be achieved using the proposed software. More interestingly, the meshfree method remarkably outperforms the FEM solution (please see Table 8). This quantification procedure could also be iterated over the entire chip root. Nonetheless, only the last 6 teeth are selected for the comparison purpose since the same pattern is more or less repeated throughout the chip length.

The last comparison in this regard is dedicated to the chip thickness. For this purpose, both minimum and maximum values of the chip thickness are reported for the simulation and experimental results. Table 9 summarizes these measurements for the present meshfree method versus the reference FEM and experimental data by Sima and Özel [34]. In

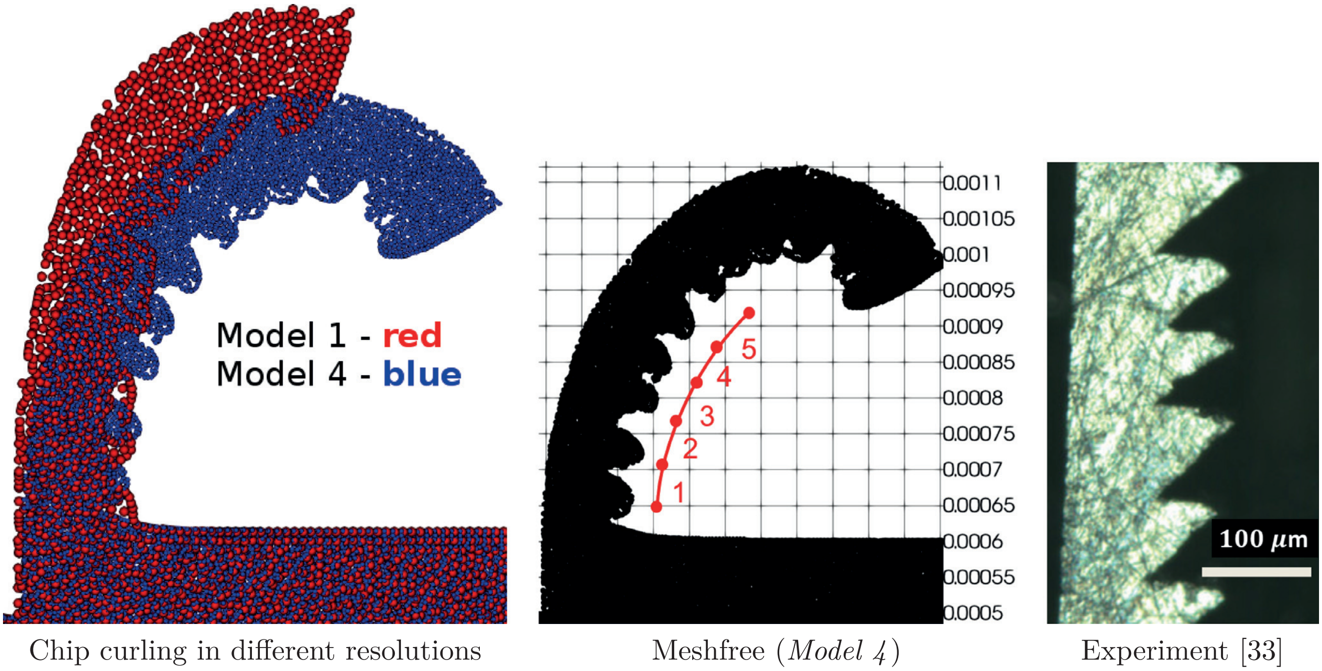

Fig. 15. Simulated chip compared to experiment with v_c = 241 m/min, r_c = 0.005 mm, and d = 0.1 mm.

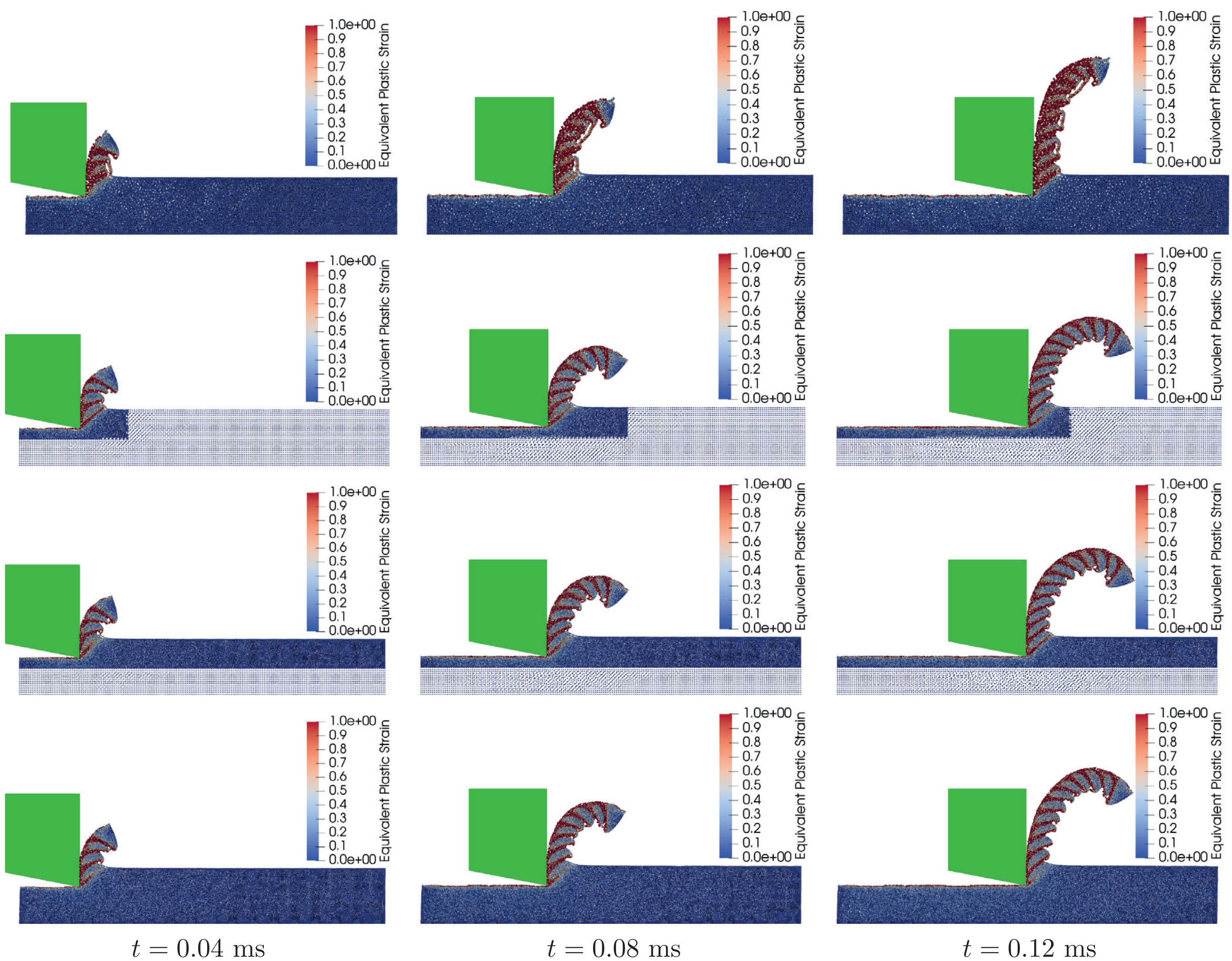

Fig. 16. Snapshots of equivalent plastic strain field, limited to 100%. From top to bottom: *Model 1* to *Model 4*, respectively.

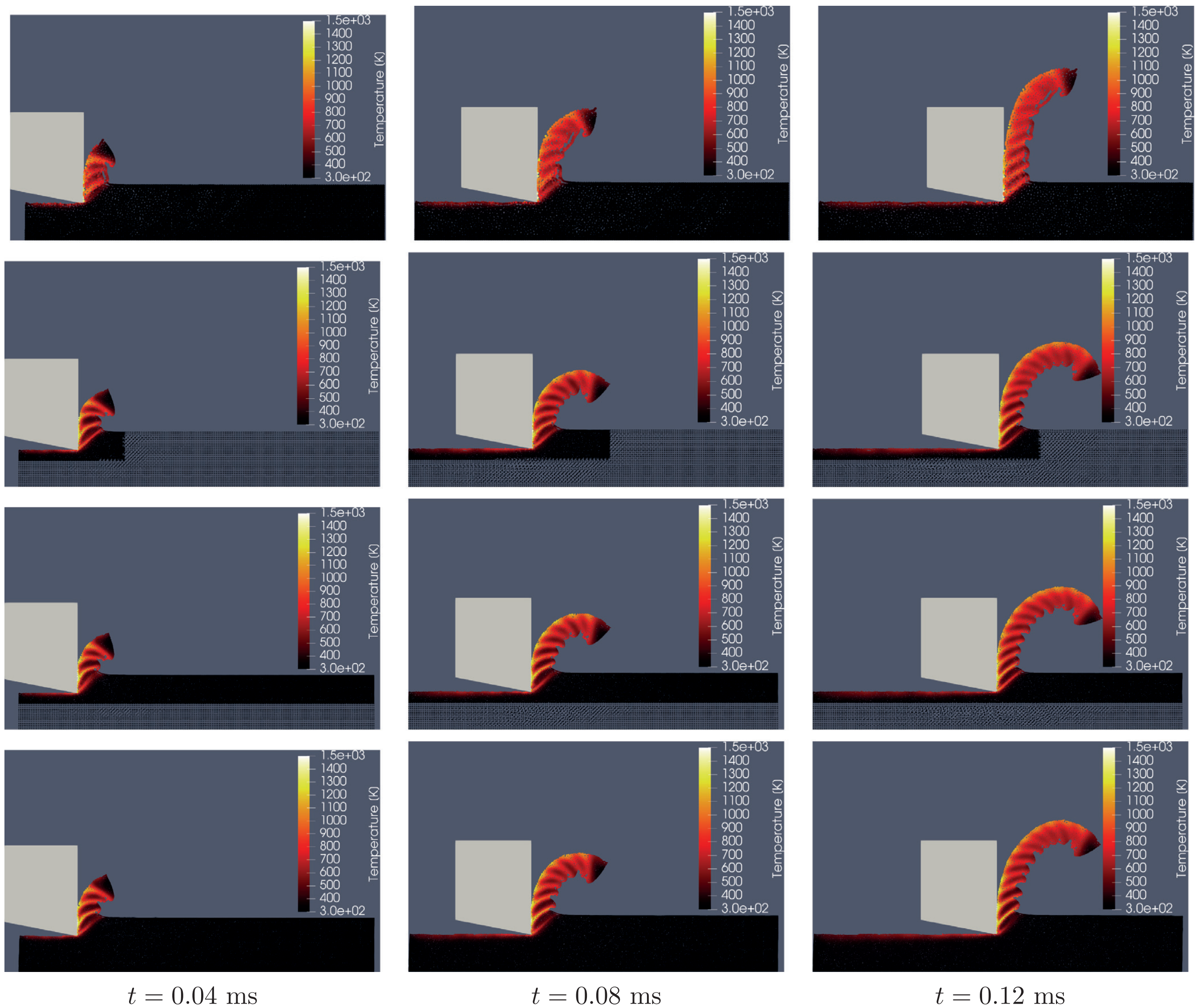

Fig. 17. Snapshots of temperature distribution. From top to bottom: *Model 1* to *Model 4*, respectively.

Table 9
Chip thickness measured by three different methods for the cutting with v_c = 241 m/min, r_c = 0.005 mm, and d = 0.1 mm.

Component	Meshfree	FEM [34]	Experiment [34]
$\tau_{\min}$ [μm]	84	107	87
$\tau_{\max}$ [μm]	140	137	160

Table 9, $\tau_{\min}$ and $\tau_{\max}$ represent the minimum and maximum thickness, respectively.

6.2. Qualitative assessment

The equivalent plastic strain is first demonstrated in Fig. 16 to acknowledge the functionality of the MJC flow stress. Claimed to be capable of capturing the thermal softening phenomenon [34], the result in fact reveals the incidence of adiabatic shear bands due to this effect, leading to a serrated chip shape. The strain localization as well is clearly visible in this figure. The equivalent plastic strain in Fig. 16 is limited to 100% for a better visualization of multiple shear bands in the generated chip. As discussed earlier in Section 1, the chip curl and consistency issue of the uncorrected SPH kernels are partially addressed by increasing the resolution. This point can be spotted by comparing the chip form of *Model 1* with those of the other three models in Fig. 16. Even using a low resolution model without any refinement, the proposed meshfree solver seems capable of retrieving the chip curl to some extent (c.f., the first row from top in Fig. 16), exhibiting the strength of the stabilized algorithm presented.

Next, it is intended to gain some insight into the functionality of the thermo-mechanical coupling by focusing on the thermal conduction effect. Towards this end, the temperature of particles is displayed in Fig. 17 for each model. In this figure, the snapshots associated with the same three time instants as Fig. 16 are shown, wherein the color represents the temperature in °K. Similar to Fig. 16, the results obtained from *Model 2* to *Model 4* are almost identical, whereas the existing deviation in *Model 1* is pertinent to its 50% lower resolution.

Figs. 17 and 18 permits several observations. First of all, the conducting trend of the transient heat introduced by the plastic work can be noticed in all simulations. The heat is spread out from the shear bands into the chip, making the surrounding of the shear bands hotter. To the best the authors' knowledge, there is no experimental data available for validating the numerical values of the temperature in this

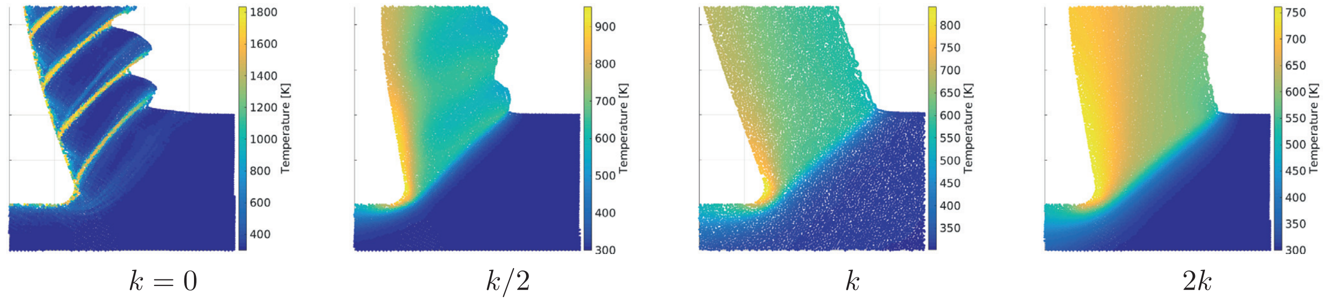

Fig. 18. Close-up of the chip root demonstrating the impact of heat conduction with the thermal conductivity k on the serrated chip formation when using the standard JC model and150,000 particles, according to [14].

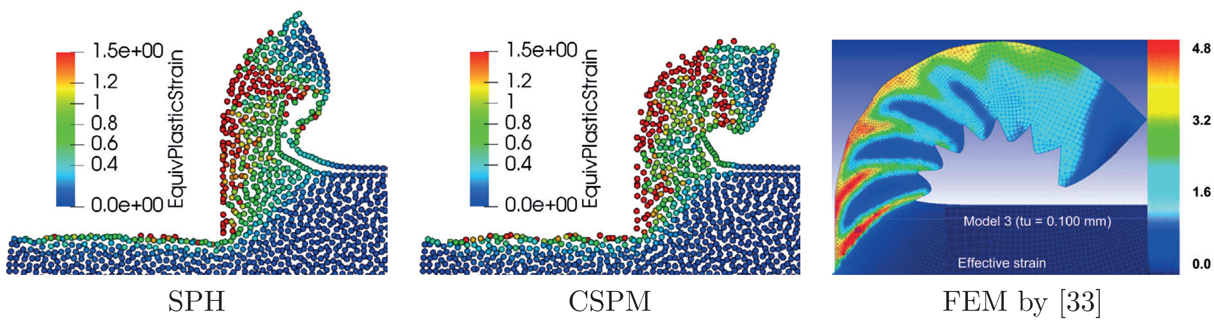

Fig. 19. Example frames of the meshfree schemes for this cutting application compared to the FEM solution with v_c = 120 m/min. A low-resolution simulation using only 30 particles along the height of the workpiece is considered on purpose to highlight the robustness of the method. The color in the SPH and CSPM plots represents the equivalent plastic strain. Although being a computationally more expensive solution, CSPM reveals the ability to reproduce a better chip curling compared to SPH due to its corrected kernel and higher consistency.

Table 10
Overview of the present metal cutting simulations.

Simulation	N_{start}	N_{end}	Runtime [min]	#Adiabatic shear bands	Correct chip morphology
Model 1	6231	6231	59	10	×
Model 2	6807	11,953	182	12	✓
Model 3	15,847	15,847	362	12	✓
Model 4	24,461	24,461	540	12	✓

operation. So, any more detailed comparison between the present temperature distribution and other references is left out of the current discussion.

Secondly, the obtained results do not overlook the serrated pattern in the chip formation even with activating the heat conduction. It is noted that employing the MJC model with extra term for strain softening is the source of this enhancement. This matter was first propounded by Roethlin et al. [14] using yet another modified version of the JC model. It was shown in their work that if the standard JC model is used, the chip serration can only be captured at very high resolutions and if the thermal conduction is completely deactivated. As demonstrated in Fig. 18, uncoupling the thermal part from the model would lead to unrealistic result for the cutting of TiAl6V4 with almost 100% deviation of the maximum temperature.

Lastly, the PSE scheme adopted shows excellent performance in solving the heat equation. It is obvious from the plots given in Fig. 17 that the adiabatic boundary conditions are handled properly with PSE.

To accentuate the impact of higher order consistent schemes on quality of results, *Model 4* is re-run using CSPM. The simulation continues until t = 0.034 ms, when the amount of chip curl leads to a complete overturning situation. The comparison of the same sequence in both SPH and CSPM, plus the FEM result of [34] are included in Fig. 19. As expected, a higher order approximation performed by CSPM can deliver a more accurate solution for the chip curl. Note that the FEM solution is obtained for a longer processing time, hence more serrated chip segments vs. the meshfree results. While vividly highlighting the robustness of the proposed solver, it looks notoriously difficult to draw any conclusive remark except the amount of chip curling studied experimentally by Sima and Özel [34].

It is understood from the comparisons presented in Figs. 19 and 15 that a realistic chip reproduction can be expected if the meshfree scheme is linear complete (i.e., first-order consistent like CSPM) and/or the resolution is high enough. While *Model 1* clearly fails to reproduce the chip morphology appropriately due to its low resolution and lack of higher order consistency, the other 3 (high resolution) models can all produce results comparable with the available experimental reference. In summary, *Model 2* is able to reach the same level of results quality in this test by circumventing almost 70% and 50% of the computational costs of *Model 4* and *Model 3*, respectively.

Finally, the measured CPU runtimes of the simulations implemented in C+ +14 are reported, taken on a single core of Intel® Core^{TM} i5-4690 at 3.50 GHz. To compare the computational efforts corresponding to each simulation, an overview table for the test case of v_c = 500 m/min is given in Table 10.

7. Conclusions and future work

As a groundwork towards optimal adaptivity in meshfree metal cutting simulations, a dynamic refinement approach was implemented to leverage the available in-house code. The main contribution of this work thus lies in an effective tailoring of the spatial refinement in the updated Lagrangian frame, tuned for an orthogonal metal cutting problem. Through a preliminary study, 4 refinement arrangements (i.e., triangular, cubic, extended cubic, and hexagonal) were first cross-compared in a density approximation error analysis. A uniform mass distribution was used to ensure the mass and linear momentum conservations. Motivated by delivering lower error norms using less particles, the cubic pattern was consequently selected and employed for the cutting simulation in Section 6. Among various possible criteria, a moving refinement frame was chosen as the criterion for splitting particles. This straightforward choice proved well suited for the application at hand.

Furthermore, the MJC constitutive law with the thermal softening term [34] was used to give rise to segmented chips of TiAl6V4. The reliability of the present approach is declared by comparing the results against available FEM/experimental data. Last but not least, the refinement algorithm allows for high resolution simulations in a more reasonable amount of time, saving almost 70% of the computational effort. As a result of this research, 1 mm of (fairly) high resolution metal cutting can now be simulated in about 3 h ending with roughly 12,000 particles.

Future work could fall into the following three categories.

1. *Maturity of the physical model*: utilizing a more advanced friction model is propounded, especially if the thermal aspect is of more interest. The appearance of new material parameters in modified JC models stresses the need for new experimental studies. Further research is indeed required to identify these parameters, perhaps with the help of inverse fitting techniques and optimizing simulations. This is due to the fact that metal cutting can easily expose deformation rates in the order of 10^6 1/s which is practically impossible to reach via existing material testing procedures, including SHPB techniques. More realistic thermal boundary conditions might be another opportunity to point out. Although the simulation results presented are in acceptable agreement with experiments, consideration of the heat transfer from the workpiece to the tool as well as the frictional heat would be necessary for model refinement.
2. *Maturity of the numerical method*: higher order consistent kernels could be investigated more thoroughly, especially in terms of the additional cost they entail. The PSE method for discretizing the heat equation could also be replaced with contemporary schemes of higher order consistency such as the second-order schemes introduced by Fatehi et al. [76] and Korzilius et al. [77], namely FNMS and ICSPM, respectively. The proposed meshfree solver could be coupled with a separate geometrical tool like the family of the Level Set methods [78] for capturing chip forms more precisely.
3. *Efficiency of the computational performance*: a prompt follow-up work could apply the coarsening procedure as well, to those particles leaving the refinement zone, for instance. This may, in turn, pave the road towards an ultimately adaptive scheme when both splitting and coalescing are enabled. Assessing appropriate adaptivity criteria together with a consistent variable-resolution algorithm must be taken into account before extending to 3D. Meanwhile, parallelizing the proposed code on the GPU could reap the benefit of tremendous computational resources offered.

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References

[1] Gingold RA, Monaghan JJ. Smoothed particle hydrodynamics: theory and application to non-spherical stars. Mon Not Royal Astron Soc 1977;181(3):375–89.
[2] Lucy LB. A numerical approach to the testing of the fission hypothesis. Astron J 1977;82:1013–24.
[3] Liu W, Jun S, Li S, Adee J, Belyschko T. Reproducing kernel particle for structural dynamics. Int J Numer Methods Fluid 1995;38:1655–79.
[4] Rabczuk T, Belytschko T. Adaptivity for structured meshfree particle methods in 2d and 3d. Int J Num Method Eng 2005;63(11):1559–82.
[5] Morris JP. Simulating surface tension with smoothed particle hydrodynamics. Int J Num Method Fluid 2000;33(3):333–53.
[6] Hu X, Adams NA. A multi-phase sph method for macroscopic and mesoscopic flows. J Comput Phys 2006;213(2):844–61.
[7] Afrasiabi M, Mohammadi S. Analysis of bubble pulsations of underwater explosions by the smoothed particle hydrodynamics method. In: ECCOMAS international conference on particle based methods, Spain; 2009.
[8] Basa M, Quinlan NJ, Lastiwka M. Robustness and accuracy of sph formulations for viscous flow. Int J Num Method Fluid 2009;60(10):1127–48.
[9] Afrasiabi M, Roethlin M, Wegener K. Thermal simulation in multiphase incompressible flows using coupled meshfree and particle level set methods. Comput Method Appl Mech Eng 2018;336:667–94.
[10] Afrasiabi M, Roethlin M, Chatzi E, Wegener K. A robust particle-based solver for modeling heat transfer in multiphase flows. In: 6th European Conference on Computational Mechanics (ECCM-ECFD), UK; 2018. p. 1–12.
[11] Rüttimann N. Simulation of metal cutting processes using meshfree methods. Diss., Eidgenössische Technische Hochschule ETH Zürich, Nr. 20646; 2012. Ph.D. thesis.
[12] Niu W, Mo R, Liu G, Sun H, Dong X, Wang G. Modeling of orthogonal cutting process of a2024-t351 with an improved sph method. Int J Advan Manuf Technol 2018;95(1–4):905–19.
[13] Niu W, Mo R, Sun H, Gnanasekaran B, Zhu Y, Liu G. A smooth particle hydrodynamic model for two-dimensional numerical simulation of ti–6al–4v serrated chip deformation based on tanh constitutive law. In: Proceedings of the Institution of Mechanical Engineers, Part C: Journal of mechanical engineering science; 2019. 0954406218803981
[14] Röthlin M, Klippel H, Afrasiabi M, Wegener K. Metal cutting simulations using smoothed particle hydrodynamics on the GPU. Int J Advan Manuf Technol 2019;102(9–12):3445–57.
[15] Bil H, Kılıç SE, Tekkaya AE. A comparison of orthogonal cutting data from experiments with three different finite element models. Int J Mach Tool Manuf 2004;44(9):933–44.
[16] Ceretti E, Lucchi M, Altan T. Fem simulation of orthogonal cutting: serrated chip formation. J Mater Proc Technol 1999;95(1–3):17–26.
[17] Calamaz M, Coupard D, Girot F. A new material model for 2d numerical simulation of serrated chip formation when machining titanium alloy ti–6al–4v. Int J Mach Tool Manuf 2008;48(3–4):275–88.
[18] Børve S, Omang M, Trulsen J. Regularized smoothed particle hydrodynamics with improved multi-resolution handling. J Comput Phys 2005;208(1):345–67.
[19] Lastiwka M, Quinlan N, Basa M. Adaptive particle distribution for smoothed particle hydrodynamics. Int J Num Method Fluid 2005;47(10–11):1403–9.
[20] Cottet G-H, Koumoutsakos PD. Vortex methods: theory and practice. Cambridge university press; 2000.
[21] Bergdorf M, Cottet G-H, Koumoutsakos P. Multilevel adaptive particle methods for convection-diffusion equations. Multiscale Model Simul 2005;4(1):328–57.
[22] Reboux S, Schrader B, Sbalzarini IF. A self-organizing lagrangian particle method for adaptive-resolution advection–diffusion simulations. J Comput Phys 2012;231(9):3623–46.
[23] Koumoutsakos P. Multiscale flow simulations using particles. Annu Rev Fluid Mech 2005;37:457–87.
[24] Feldman J, Bonet J. Dynamic refinement and boundary contact forces in sph with applications in fluid flow problems. Int J Num Method Eng 2007;72(3):295–324.
[25] López YR, Roose D, Morfa CR. Dynamic particle refinement in sph: application to free surface flow and non-cohesive soil simulations. Comput Mech 2013;51(5):731–41.
[26] Vacondio R, Rogers B, Stansby P, Mignosa P, Feldman J. Variable resolution for sph: a dynamic particle coalescing and splitting scheme. Comput Method Appl Mech Eng 2013;256:132–48.
[27] Barcarolo DA, Le Touzé D, Oger G, De Vuyst F. Adaptive particle refinement and derefinement applied to the smoothed particle hydrodynamics method. J Comput Phys 2014;273:640–57.
[28] Hu W, Pan W, Rakhsha M, Tian Q, Hu H, Negrut D. A consistent multi-resolution smoothed particle hydrodynamics method. Comput Method Appl Mech Eng 2017;324:278–99.
[29] Rabczuk T, Belytschko T. A three-dimensional large deformation meshfree method for arbitrary evolving cracks. Comput Method Appl Mech Eng 2007;196(29):2777–99.
[30] Vacondio R, Rogers B, Stansby P, Mignosa P. Variable resolution for sph in three dimensions: towards optimal splitting and coalescing for dynamic adaptivity. Comput Method Appl Mech Eng 2016;300:442–60.
[31] Rabczuk T, Samaniego E. Discontinuous modelling of shear bands using adaptive meshfree methods. Comput Method Appl Mech Eng 2008;197(6):641–58.
[32] Spreng F, Schnabel D, Mueller A, Eberhard P. A local adaptive discretization algorithm for smoothed particle hydrodynamics. Comput Part Mech 2014;1(2):131–45.
[33] Spreng F, Eberhard P. Machining process simulations with smoothed particle hydrodynamics. Procedia CIRP 2015;31:94–9.
[34] Sima M, Özel T. Modified material constitutive models for serrated chip formation simulations and experimental validation in machining of titanium alloy ti–6al–4v. Int J Mach Tool Manuf 2010;50(11):943–60.
[35] Jaspers S, Dautzenberg J. Material behaviour in metal cutting: strains, strain rates and temperatures in chip formation. J Mater Proc Technol 2002;121(1):123–35.
[36] Boothroyd G. Fundamentals of metal machining and machine tools, 28. Crc Press; 1988.

[37] Shaw MC. Metal cutting principles, 2. Oxford university press New York; 2005.
[38] Wyen C-F. Rounded cutting edges and their influence in machining titanium. ETH Zurich; 2011. Ph.D. thesis.
[39] Tönshoff HK. Spanen: Grundlagen. Springer-Verlag; 2013.
[40] Belytschko T, Liu WK, Moran B, Elkhodary K. Nonlinear finite elements for continua and structures. John wiley & sons; 2013.
[41] Taylor GI, Quinney H. The latent energy remaining in a metal after cold working. Proc R Soc Lond A 1934;143(849):307–26.
[42] Swegle J, Attaway S, Heinstein M, Mello F, Hicks D. An analysis of smoothed particle hydrodynamics. Tech Rep. Sandia National Labs., Albuquerque, NM (United States); 1994.
[43] Li G, Belytschko T. Element-free galerkin method for contact problems in metal forming analysis. Eng Comput 2001;18(1/2):62–78.
[44] Vignjevic R, De Vuyst T, Campbell J, Source C. A frictionless contact algorithm for meshless methods. Comput Model Eng Sci 2006;13(1):35.
[45] Röthlin M. Meshless Software Tool to Simulate Metal Cutting Operations by Employing Contemporary Numerical Methods. PhD diss. ETH Zurich; 2019. ISBN 978-3-906916-51-4.
[46] Libersky LD, Petschek A. Smooth particle hydrodynamics with strength of materials. In: Advances in the free-Lagrange method including contributions on adaptive gridding and the smooth particle hydrodynamics method. Springer; 1991. p. 248–57.
[47] Jaumann G. Geschlossenes system physicalisher und chemischer differentialgesetze. Sitzber Akad Wiss Wien (IIa) 1911;120:385–530.
[48] Johnson GR. A constitutive model and data for materials subjected to large strains, high strain rates, and high temperatures. Proc 7th Inf Sympo Ballistics 1983:541–7.
[49] Lee W-S, Lin C-F. Plastic deformation and fracture behaviour of ti–6al–4v alloy loaded with high strain rate under various temperatures. Mater Sci Eng A 1998;241(1–2):48–59.
[50] Monaghan JJ. Simulating free surface flows with sph. J Comput Phys 1994;110(2):399–406.
[51] Monaghan JJ. Smoothed particle hydrodynamics. Rep Prog Phys 2005;68(8):1703.
[52] Liu G-R, Liu MB. Smoothed particle hydrodynamics: a meshfree particle method. World Sci; 2003.
[53] Li S, Liu WK. Meshfree particle methods. Springer Science & Business Media; 2007.
[54] Deligonul Z, Bilgen S. Solution of the volterra equation of renewal theory with the galerkin technique using cubic splines. J Stat Comput Simul 1984;20(1):37–45.
[55] Li S, Liu WK. Meshfree and particle methods and their applications. Appl Mech Rev 2002;55(1):1–34.
[56] Eldredge JD, Leonard A, Colonius T. A general deterministic treatment of derivatives in particle methods. J Comput Phys 2002;180(2):686–709.
[57] Fatehi R, Fayazbakhsh M, Manzari M. On discretization of second-order derivatives in smoothed particle hydrodynamics. In: Proceedings of World Academy of Science, Engineering and Technology, 30. Citeseer; 2008. p. 243–6.
[58] Mas-Gallic S, Raviart P. Particle approximation of convection-diffusion problems. Report of Univ de Paris, 6; 1986.
[59] Choquin J, Huberson S. Particles simulation of viscous flow. Comput Fluid 1989;17(2):397–410.
[60] Afrasiabi M, Roethlin M, Wegener K. Contemporary meshfree methods for three dimensional heat conduction problems. Under Peer Rev 2019.
[61] Randles P, Libersky L. Smoothed particle hydrodynamics: some recent improvements and applications. Comput Method Appl Mech Eng 1996;139(1):375–408.
[62] Chen J, Beraun J, Jih C. An improvement for tensile instability in smoothed particle hydrodynamics. Comput Mech 1999;23(4):279–87.
[63] Monaghan J, Gingold R. Shock simulation by the particle method sph. J Comput Phys 1983;52(2):374–89.
[64] Monaghan JJ. Smoothed particle hydrodynamics. Ann Rev Astron Astrophys 1992;30(1):543–74.
[65] Gray J, Monaghan J, Swift R. Sph elastic dynamics. Comput Method Appl Mech Eng 2001;190(49–50):6641–62.
[66] Adami S, Hu X, Adams NA. A transport-velocity formulation for smoothed particle hydrodynamics. J Comput Phys 2013;241:292–307.
[67] Monaghan J. On the problem of penetration in particle methods. J Comput Phys 1989;82(1):1–15.
[68] Klemp JB, Wilhelmson RB. The simulation of three-dimensional convective storm dynamics. J Atmosph Sci 1978;35(6):1070–96.
[69] Skeel RD. Variable step size destabilizes the störmer/leapfrog/verlet method. BIT Num Math 1993;33(1):172–5.
[70] Bentley JL. Multidimensional binary search trees used for associative searching. Commun ACM 1975;18(9):509–17.
[71] Hockney RW, Eastwood JW. Computer simulation using particles. crc Press; 1988.
[72] Feldman J. Dynamic refinement and boundary contact forces in Smoothed Particle Hydrodynamics with applications in fluid flow problems. United Kingdom: PhD Thesis, Swansea University; 2006.
[73] Afrasiabi M, Chatzi E, Wegener K. A particle strength exchange method for metal removal in laser drilling. Procedia CIRP 2018;72:1548–53.
[74] Roethlin M, Klippel H, Afrasiabi M, Wegener K. Meshless single grain cutting simulations on the GPU. Int J Mechatr Manuf Syst 2019. Accepted.
[75] Jawahir I, Brinksmeier E, M'saoubi R, Aspinwall D, Outeiro J, Meyer D, et al. Surface integrity in material removal processes: recent advances. CIRP Ann-Manuf Technol 2011;60(2):603–26.
[76] Fatehi R, Manzari M. Error estimation in smoothed particle hydrodynamics and a new scheme for second derivatives. Comput Math Applica 2011;61(2):482–98.
[77] Korzilius S, Schilders W, Anthonissen M. An improved cspm approach for accurate second-derivative approximations with sph. J Appl Math Phys 2016;5(01):168.
[78] Osher S, Fedkiw R. Level set methods and dynamic implicit surfaces, 153. Springer Science & Business Media; 2006.